



A BRIEF SYNOPSIS

We are R&D Conglomeration with successful validation of our medicines by extensive clinical trials and establishing our credentials. Some of our innovation were published worldwide and highly acclaimed during conferences globally. Below is a very short description, preparations and methods. All these are successfully tried on about 15000 volunteers almost in many countries including America, Europe, China and Middle east.

WWMA AG group main scientific discoveries and generated medications

For more than 30 years a group of scientific researchers associated under Longevity World Holding with a head company World Wide Medical Assistance AG (WWMAAG) executes an active scientific research in a pharmaceutical, medical and veterinary field. A huge experience in practice and relevant information and statistical data base (more than 1 million clinical cases with laboratory and farm animals and about 5 thousand of in-vivo researches in the field of a human medicine), as well as a deep knowledge of the fundamental biological mechanisms and creativity allowed us to pass a long way from the diagnosis and infectious diseases treatment of various veterinary aetiology to the perfect method of early oncological disease detection and to the methodology that allows to create a line of medications based on the active substances treatment in high dilution (HD).

Taken following thesis as the basic: **“Active longevity means an assured absence of oncological diseases and efficient prophylaxis of metabolic and infectious diseases”** – the WWMA AG scientists elaborated a **system of measures** including medications and the methods of their use that allows under control **to guarantee active longevity till 90-100 years old to everybody** without hard and chronic diseases.

The major discovery made within scientific and research laboratories of WWMA AG company more than 20 years ago is elaboration of methods and diagnosticums for **detection of biochemical, bioenergetical and respectively pharmacological activity of natural and synthesized substances in HD**, i.e. well-defined evaluation of a “top of activity” and it’s lowermost and higher margins.

This discovery and cumulated experience of preclinical researches (cells culture, chicken embryos, genetics, farm and lab animals) allowed to create several stocks of pharmacological medications and organize scientific and clinical research of those medications with respect to 20 nozologies) of basic diseases of XX-XXI cent.

Few of our uniqueness includes:

- the largest volume of research performed on farm and laboratory animals - more than five million cases;
- fundamental research of bioorganisms homeostasis and redox of the system that manages them;
- developed on the basis of the conducted researches of innovative technologies of production of fundamentally new drugs;
- successful clinical studies and years of application period established drugs for the prevention and treatment of patients-volunteers;



Basic are the following technologies:

1. biological reactor - bio-model on the basis of a unique cultivated (BTIH Singapore and WWMA AG) strain of the bacteria Esherhi-Koli used to identify new properties of synthesized and natural substances in ultra-low (high dilution till M-16) doses (properties are typical only in microcosm);

2. technology, allowing the use of low molecular weight proteins for the production of the preparations restoring the functional activity of the endocrine system and allowing for prevention and therapy of pancreas and thyroid gland, prostate and other organs and functional systems;

3. the third technology is the methods of using products of microorganisms in drug development that allows to produce stability of bioorganisms (animals and humans) infectious diseases through the mechanism of «specific immunity» with a complete lifetime embodiment, fixing it in immunocompetent organs, and for individual representatives of bioorganisms - transfer mechanism of this resistance in generations (maternal line);

In all these technologies there are patents or patent prepared formula.

4. Through the use of endogenous and secreted components of microorganisms established antiviral drugs for the treatment of human and animal viral infections (the drugs effect on waste products and by modifying their properties, allow their use as therapeutic agents in the internal environment of the organism, the so-called endogenous vaccines and pharmacological agents). This creates long-term protection of bioorganism which forms the basis of endogenous (autologous) vaccines, regardless of the type of pathogen.

5. WWMA AG developed technology and equipment for production of fundamentally new substances, which secure:

- selective elimination of «lower « bio-organisms: bacteria, viruses, fungi; apoptosis of cells, infected by pathogenic factors within the body.

Developed appropriate compositions for the production of new substances for different purposes:

- pharmacological preparations for the prevention and treatment of bio-organisms (human and animal) for internal and external use;

- antibacterial drugs for elimination of pathogenic factors of bacterial, viral and fungal etiology in premises (housing , farms, medical facilities), for disinfection in public areas, private accommodation, meals, prisons etc.

Such kind of drugs were created by us for the prevention and treatment of bioorganisms from bacterial and fungal infections, when this mechanism was involved in the maintenance of the microbiota (micro-biocenosis in the intestine, in the blood).

The main areas of work are as follows:

- Development of methods of early diagnostics and prevention of a number of grave infectious and degenerative-type diseases;

- Development of medicines and methods of their use for curing diseases associated with metabolic syndrome and aging of the human system, as well as of infectious pathologies of bacterial, viral and fungous aetiology;

- Development of medicines and methods of curing neurodegenerative-type diseases which are developed in the late ontogenesis: Alzheimer's disease, Parkinson's disease, dementia and others etc.;

- Development of "human system's homeostasis tuning system" on the basis of use of the lines of creating medicines and geroprotectors, with adjusting it to physiological balance of organism homeostasis.



An important task of the WWMA AG scientific program of restoration of homeostasis deviations - is the research of properties of a line of medicines created on the basis of natural metabolites and synthetic substance, used in the programs of prevention and cure of general obesity, abdominal obesity and the associated metabolic syndrome.

The scientific potential and new technological possibilities permitted the research team of WWMA AG and are Russian branch "NIMB Ltd" to create essentially new methodologies and medicines for prevention, cure and rehabilitation of a number of diseases of the human system. One of the main substances used for creating medicines is a natural (endogenous) human system metabolite (formaldehyde). It is generated under the influence of liver microsomal ferments and interacts with amines, amides, proteins, nucleoproteids, nucleosides, nucleic acids. Actually the basis of life were laid in the metabolic tracts of homeostasis, which involve the molecules of methyl group, **as well as aldehyde of formic acid and formiate as products of oxidation of these methyl groups.**

To it testify also discovered archaic methanotrophic bacteria, which use methane in their vital activity as the only source of carbon and energy, confirm our hypothesis. Methane inside a cell is oxidized according to the chain "methane – methanol – formaldehyde – formiate". This metabolic junction is actually the regulator of main metabolic flows and the basic oxidation-reduction balance, i.e. the basis of life.



Basic principles for medications to be generated:

- scientific hypotheses being advanced by the team of scientists had got proof after laboratory and clinic research with replicated results obtained in independent laboratories. Scientific team of the group of the companies elaborating such medications counts more than 50 specialists including academicians, doctors and candidates of science;
- efficiency of obtained medications is proved by practical work in spite of initial theoretical hypotheses run counter to conventionally accepted conceptions about properties of the range of chemical compounds and their impact on alive system;
- generated technologies for industrial production of the medications ensure high profitability of business-process with minimum expenses and secure short terms of projects payback after introduction of mentioned elaborations; effectiveness of medications and application methods appropriate to existing infrastructure of medicinal institutions and professional resources of the specialists provide successfully their progression in the market;
- expected competitive resistance in the market of pharmaceutical medications and medicinal service is surmounted thanks to introduction of elaborations which don't have analogs over the world and which substitute severe (with invasions) and / or ineffective methods of treatment.
- up-to-date scientific knowledge is surviving a deep systemic crisis stipulated by discordance of the observed effects to the fundamental theoretic constructions. This affirmation regards to pharmacology as well: rising in prices of medicinal and pharmacological elaborations by several orders of magnitude, uncontrollable enlargement of negative incidental effects, and as a result – decrease of efficiency with simultaneous boost of prices for medications and medicinal service. Among alternative versions to go out of the crisis we see retrospective analysis of theoretical constructions of the past those have been either incorrectly comprehended or merely forgotten, but nowadays we apply them with assessable relevant researching mechanism on the cell and molecular levels;
- new world outlook and new scientific ideology are working out on the base of fundamental principles discovered in the past and on the base of knowledge about natural phenomena obtained with help of contemporary scientific technologies;
- monitoring of the conceptions having summarized in the field of an ancient naturopathic, western symptomatic and ultra-modern high-tech medicine with involvement of world outlook approaches in physics, chemistry, mathematics, quantum mechanics and astrophysics allows us to conduct teleological quest of the medications and their optimal dosages to stimulate impact on such perfect homeostatic system like human organism;
- experience accumulated by us during three decades of years let us contend that the fundamental elements and systems where alive nature is based on have been discovered. They are the base of existence, functioning and reproduction for living organisms and they are universal for all alive beings. These elements are the simplest chemical compounds acting in optimal concentrations in homeostatic system. Homeostasis breakdown is a cause of living organism death. Our medications are aimed at optimization of the process of such system functioning that secures its' resistance to the exterior impacts and interior environment alterations which launch a pathological process;



Urgency and validity of the projects realization in the current time:

- the projects in the medicinal field – oncology, cardio-vascular diseases, reproduction and geroprophylaxis; in agricultural field - invigoration and enlargement livestock of farm animals and poultry population, supplementary weight gain (without hormones, antibiotics and genetic engineering use) of farm animals, poultry and fish provision, and as well in agronomy – yield increasing and grain-crops tolerance – are urgent and relevant to the priority directions of the national programmes in public healthcare and agriculture of each country over the world regions. Their urgency has particularly increased in the situation of a worldwide crisis, by virtue of the fact that expenses for projects realization are just thousandth from their economical effect, and a social effect is inestimable;

- in principle a new generation of the medications with their overall impact on a biosystem allows to significantly decrease the costs for social programmes or to more effectively distribute them with simultaneous increasing of medical efficiency to the level that have never been achieved in a contemporary society. Such concept will allow to avoid the troubles for all social layers of the population, which are about to happen due to deteriorating quality of medical service, a lack of funds for both medicine and pensions. Private companies being in possession of such methods will take the leading position in the field and will use a powerful administrative support of the government;

- elaborated methods regard first of all to the pathologies which have got the major (pandemic) world extension during 50 last years: hepatitis, allergy, clamidiosis and candidiasis, HIV, metabolic syndrome and oncology. Such projects are the priority as by definition of World Health Organization and United Nations Organization (UNO), as for the majority of countries and commercial niches of medicinal service market;

- oncological researches correspond to the plans of the world association to establish the Centers of the palliative methods for oncological diseases treatment. In our case the nature of elaborated medications and methods impact allows not just to efficiently treat oncological diseases dilating thereby the number of the patients, but as well to avoid serious incidental effects on the organism on the whole;

- conduction of clinical tests required for new pharmaceutical medications registration has been considerably simplified due to the company put into effect the complete cycle of such investigations earlier.



The research made by the corporate group «World of Longevity» leded by WWMA AG, Switzerland in biomedicine and nanopharmacology.

1. The essence of innovations.

1.1 The purpose of the creation was design of the new generation, medical equipment, drugs and methods for providing of effective human activity, also we investigated the basic mechanisms of metabolic processes and the functional activity of cells bio-organisms on the atomic and molecular level, including the retrospective analysis of the evolution of metabolism till it appeared on earth 3.6 billion years ago, which allowed us to make a number of important and fundamental discoveries.

- We have developed methods and diagnosticums to determine the areas of biochemical and pharmacological activity of natural and synthetic substances in ultra-low doses and nano-doses, which enabled the creation of a new generation of pharmaceutical products;

- We have developed the the technology for production of small molecules of a specified size, and the «content» and methods of their use for therapeutic purposes;

- We have developed the theory and studied the mechanisms of epigenetic regulation of homeostasis and function of cells of a multicellular organism by the small molecules;

- We have developed a real model for the evolution of metabolic processes in living organisms since the moment of transition from the stage of chemical evolution to biochemical: the appearence of the ancient forms of living matter - archaea, methanotrophic and methanogenic bacteria and based on it, we have drawn the scheme of the metabolic processes that maintain energy homeostasis and biochemical all functional systems of somatic cell in bio-organisms;

- We have compiled the generalized scheme of somatic cell metabolism in the human body that allows us to identify the basic metabolic pathways and metabolites, which are the markers of pathology and aging; they took their place by the evolution way and and objectively characterizing the state of biochemical processes, tissues, organs and systems at the organismic level and play the main role in these processes;

- We have developed the lipid theory of aging of the human body;

- We have made the lines of the new generation pharmacological agents that have both generalized (immunomodulatory, regenerating) and specific (preventive and curative) effects on human the body; these lines were made for the treatment of the following diseases:

- Systemic diseases;

- Diseases associated with metabolic syndrome;

- Infectious diseases of bacterial, viral and fungal etiology;

- The type of degenerative disease, including cancer various etiologies;

- Neurodegenerative diseases: dementia, Parkinson and Alzheimer's disease, multiple sclerosis and central nervous system diseases.



The formulation of pharmaceutical and cosmetic products is based on the rules for creating of ideal preparations:

- The chemical composition and structure of the product components (antigens, adjuvants, carriers, etc.) must be accurately set;
- The product should be administered for the period of time of its action;
- The product should be comprehensive and to create immunity to many infections;
- For vaccine - the vaccine must provide lifelong immunity in 100% of vaccinated;
- The preparation must be safe and has no side effects;
- The preparation should be administered by the comfort way for medical staff and patients.
- The preparation should be stable, have a long shelf -life;
- The drug should not need to comply with «cold chain»;
- The manufacturing process of the preparation must meet the upto-date requirements;
- The cost of manufacture and using of the preparation should not be high.

Nowadays, medical equipment and drugs are established, and some of which went through all kinds of research, certified and / or registered, some have international patents (up to 35 countries in all regions of the world), and most of them are used for the treatment of specific diseases.

The product having a cholesterol-regulative effect

The product having a cholesterol-regulative effect (XP / DB) and methods for its use in the treatment of diseases of the circulatory system, and diseases connected with aging. This product comprises an active part of the metabolic origin and destination additive - an isotonic sodium-chloride solution for injection 0.85-0.95% concentration.

The preparation for activating the own stem cells of the human body

The preparation to activate the stem cells (CS) for normalizing the hormonal, immunological and neurological systems of the human body containing the active metabolic origin and sodium chloride for injection, characterized that it contains a natural metabolite as active ingredient. Administered by a specific procedure, this preparation provides (endogenously) the proliferation of autologous stem cells in the body at the level of a young organism, increasing this proliferation as related to the organism in the middle and later ontogenesis: in the bone marrow - in 11-13 times, and in the spleen and the liver - in 4-5 times.

The preparation for the prevention and treatment of neurodegenerative diseases

We have designed the drugs and methods of their use for the prevention and treatment of the early and middle stages of the development of neurodegenerative diseases and for improvement of the quality of life of patients in late ontogeny: Alzheimer's disease, Parkinson's disease (tremor), hydrocyanic dementia.



The preparation for normalization of metabolic processes in the organs and tissues

The preparation normalizing the redox processes and metabolism at the cellular and organismic level (MET-GP, GP / PQ), containing chlorine-sodium compound, characterized in that it further comprises a natural metabolite, but the sodium chloride for injection (concentration of 0.85-0.95%) acts as chlorine-sodium compound. The target application of this preparation as a basic component of the line of geroprotectors ensuring prevention of violations and restoring cell metabolism, normalizing metabolic processes in human organs and tissues and restoring homeostasis disorders caused by its phenotypic alterations in late ontogeny.

The preparation for the prevention and treatment of disorders in the locomotor system.

We have designed the preparations (GGP) and methods of their use for the prevention and treatment of disorders in the locomotor system, and also for the purposes of the application of these drugs as the main components of the geroprotectors' line in the middle and late ontogeny. These preparations are used to treat arthritis, arthrosis, including deforming (prevention and treatment with early stage), and other systemic diseases of the locomotor system.

The tools and methods for complex protection of human infectious «pathogenic» factors based on the promotion of immune system in the human body, including the strengthening of T - killer component of immunity and the start of the specific complete phagocytosis.

We have developed preparations (Immuno-PNM, PNM-SMA) and a method for controlling the metabolic process and activation of the immune system in the human body: increasing T - lymphocyte (T - killer) component of the immune system, and the start of a specific complete phagocytosis for making the detoxification of the body. As active ingredients of drugs we have used the endogenous and secreted components (proteins, lipids and polysaccharides) of microorganisms, which are haptens, not having the ability to activate the production of antibodies. We have developed the mechanism and products that perpetuate the low molecular weight components of microorganisms in the protective immune organs, and the methods of their use for complex protection from infectious diseases of bacterial, viral and fungal etiology.

The administration of these components with the special procedure provides completed specific phagocytosis of pathogens (bacteria, fungi) and also the activation of the antiviral immune based on the components contained in the causative agents of viral diseases until the complete elimination of pathogens and diseases, respectively. The complete cure of the disease is stated by the lack of antibodies in the organism, which are typical of the disease.

The tools and techniques for the prevention and treatment of viral etiology infections

We have developed preparations (AGG-TEM) and the methods for prevention and treatment until complete elimination of viral etiology infections (based on the activation of the specific completed phagocytosis of viruses) including adenoviral infection, influenza, herpes.

The tools and methods for the treatment of viral hepatitis B and C.

We have developed products and methods of their use for the complete elimination of viral hepatitis B and C. The basis of the method is the epigenetic regulation of homeostasis and function of cells in the multicellular organism with the small molecules, which are produced by the basic metabolites of the cell, i.e. the basic elements of the redox system in the body.



The tools and methods for the prevention and treatment of infectious skin diseases of bacterial, viral and fungal etiology.

We have developed drugs (line «Allpharm» - liquids, gels, lotions) and methods of their use for prevention and treatment of infectious skin diseases of various etiologies, including manifestations of adenoviral infection, herpes, acne and deep mycoses, including nail.

The preparation for using against infectious diseases of viral etiology.

We have designed the unique preparations of a new generation for the prevention and treatment of serious infectious respiratory diseases and influenza (used through the atomizer). URI, influenza (all existing strains), and adeno-virus infections are eliminated. The preparation has no negative side effects, it is not a classical antibiotic, it contains no toxic elements for human body, and also is does not have resistance to all types of viruses. This preparation is indicated for patients of all ages, including young children.

The tools and methods for the complex effect on the human body (male and female) in order to prepare it for the impregnation of a healthy child, and also to restore the reproductive characteristics of the organism.

We have developed the preparation (Water-miscible concentrate - PNM) and methods for the normalization of homeostasis and metabolic processes in the body for restoring the reproductive qualities of the person, including solving the problem of male and female agenesis.

The preparations based on the natural ingredients for fungoterapia.

We have developed the products based on natural origin of funds - the Ganoderma lucidum and Cordyceps spores for complex effects on the human body for the preventive and therapeutic purposes. Unlike the similar existing preparations because of the technology of processing and release of active components, which ensure increasing effectiveness of preparations and the availability of the healing preparation by the body.

Method for instant diagnosis of various etiology cancer.

A method of qualitative instant provisional diagnosis of cancer, including blood sampling, mixing with a reagent and recording the results of reaction, characterized in supplementary preparing 0,12-0,5% citrated blood suspension and the washed erythrocytes of the same patient. A supernatant is used as a reagent and it is obtained by the cultivation of the cells of swine embryo nephros on the continuous line in anaerobic conditions, mixing with a reagent the citrated blood suspension and the washed erythrocytes suspension separately and then maintaining the mixture at a temperature of 4 to 8 ° C for 12-15 hours. Then determined separately characters of citrated blood reactions and washed erythrocytes with the reagent, and the presence of cancer is defined by the positive reaction of erythrocytes and the negative or positive reactions of citrated blood, and also the negative reaction of erythrocytes and the positive reaction of citrated blood.

The preparation for the prevention and treatment of cancer of any etiology and localization and also the different stages of the disease.

We have developed the preparations (Onco-PNM) and the methodology of an integrated approach to prevention and treatment of oncological diseases.



Preparations based on chlorine-sodium compound additionally contain the natural metabolites: human body, fungus (Ling Zhi and Cordyceps), as well as the endogenous and secreted components – haptens, which are involved by the cancer cells during their life. Transport to immunocompetent organs and consolidation of preparations in them is provided with a specially developed biochemical solutions, which are one of the main regulators of cellular metabolism.

The preparation for elimination of papillomas, keratomas and for the treatment of melanoma.

The preparation «NANO-U» for surface treatment «heads» of papillomas, keratomas, as well as the complete elimination of melanoma with the complete damage of these pathological elements' mycelium.

The natural formulations for transdermal application.

Because of the compositions of natural ingredients and technology of their transfer in «small molecule» is it possible to create a new class of I patches for the local transdermal input of drugs for pain control, muscle strain relieving, and treatment of the joint and locomotor system's diseases.

The tools and methods for the treatment of «Acquired Immune Deficiency Syndrome» (AIDS) by the method of transfer from HIVpatient stage to the HIV-infected stage.

We have developed the products and methods of their use for the treatment of «Acquired Immune Deficiency Syndrome» (AIDS) by the method of transfer from HIV- patients stage to the stage of HIV-infected on the basis of viral load decrease in the intercellular and intracellular spaces in the presence of HIV and the further maintaining of this state in the phenotype.

Summary: «Description of a number of preparations developed by WWMA AG and used for the treatment of a complex and specific therapies»:

1 Preparation XR / DB - drug having generalized and specific impact for preventive, curative and rehabilitative impact on the patient with systemic pathologies and degenerative diseases (neurodegenerative, disorders of the endocrine and hormonal systems, cancer), and age-associated diseases (geriatrics, geroprophyllaxis) .

Packing -5 ml, 100 ml, 200 ml-1, 20 and 40 injections . Drug administration - 5 ml each injection, intramuscularly. Instructions for setting an injection **)

2. Preparation CS – is for the proliferation of the genuine stem cells and prevention, treatment and rehabilitation of the patient at the final stages of complex (software) treatment of systemic diseases.

Packing -5 ml, 100 ml, 200 ml-1, 20 and 40, injections. Drug administration -5 ml each injection, intramuscularly. Instructions for setting an injection **)

3. PREPARATION GP / PQ - for prevention and treatment of diseases of the cardiovascular system: hypertension holesterinomii, hyperlipidemia, metabolic syndrome, obesity, and type II diabetes.

Packing - 10 ml, 100 ml, 200 ml of 5, 50 and 100 injections. Drug administration - 2 ml each injection, intramuscularly. Instructions for setting an injection **)



4. PREPARATION GGP - therapy and prevention of diseases of the locomotor system, including articular arthritis, arthrosis,(including deforming), osteochondrosis and locomotor pathologies: vertebral hernia, ankylosing spondylitis at the initial stage, sarcopenia, and others.

Packing - 3 ml, 100 ml, 200 ml, of 1, 33 and 66 injections. The administration of the drug - 3 ml each injection, intramuscularly. Instructions for setting an injection **)

5 PREPARATION GP - repairing and optimizing of metabolism at the cellular level, the prevention and treatment of a number of systemic diseases (in combination with other TEM-preparations) - CVD, gastrointestinal tract, the SLM, BLZ, for post-procedury recovery

Packing - 3 ml, 100 ml, 200 ml, of 1, 33 and 66 injections. The administration of the drug -3 ml each injection, intramuscularly. INSTRUCTIONS for setting an injection **)

6. Preparations of the following lines (issued in accordance with indications by the results of the patient's diagnosis)

6.1 Treatment of viral infections - infections of various etiologies: hepatitis A, B, C, etc., herpes infection, HPV infection chlamydioses, cytomegalovirus infection, influenza, ARVI, uro-genital infection localization, treatment of HIVinfection, and treatment of tick-borne encephalitis etc.

6.2 Cancer Therapy - oncology of any etiology and any stage - except ascites: 1 and 2 stages is treatment - 100/95% , 3rd Stage is recovery from 60% to 85% depending on the etiology, the term of disease and the conducted therapy, stage 4 - from 30 to 50% depending on the etiology, the term of disease and the conducted therapy;

6.3 Therapy of the endocrine system - the restoration of optimal functioning of the endocrine glands (restoring of hormonal balance) - to ensure the humoral regulation of the functions of the human body (HB) to maintain a constant of internal environment of HB (homeostasis), the regulation of reproductive function, for regulation of energy metabolism, the normalization of the psychoemotional and mental state of the patient.

6.4 Therapy of the digestive system - the whole complex of functionally related organs that provide mechanical and chemical handling of food, absorption of nutrients into the blood and lymphatic channel, the formation of stools and their excretion from the body. Recovery of the optimal regulation of body's system to ensure the energy and plastic substances; the optimization of metabolism at the cellular level, allowing the body to reduce the amount of consumed food without energy loss; restoration of the properties of maximum detoxification of exogenous and endogenous toxins in the human body. Recovery of normal activities (including regeneration) of the digestive tract organs such as: salivary glands, esophagus, stomach, intestines, liver (including cirrhosis), gallbladder and pancreas, including decrease and elimination of benign diseases, diseases associated with metabolic disorders (gout, etc.).

6.5 Treatment of urinary system diseases: nephritis, chronic glomerulonephritis, nefroangiosklerosis, pyelonephritis, cystitis, kidney stone disease, chronic nephratonia, hemorrhoids, prostatitis, BPH(benign prostatic hyperplasia).

6.6 For the prevention and treatment of the full range of age-associated diseases:



neurodegenerative, sarcopenia, osteopenia (osteoporosis, adult rickets), impotence, cardiovascular system, disorders of sleep and appetite, stress, depression and asthenic syndrome, improvement of the body's energy and improvement of the quality of life for patients in the late ontogeny (elderly and old age).

6.7 For the maintenance of the parents' bodies in order to prepare them for the impregnation of a healthy child - cure diseases, including chronic, systemic functional recovery of bodies, restore function of reproductive organs (treatment of impotence and frigidity) and the quality of semen and also menstrual disorders, menopausal syndrome and fibrocystic diseases.

6.8 Treatment of the cardiovascular system: ischemic heart disease, cardiac angina, hypertension, cerebrovascular disease, rehabilitation of post-infarcted and post-apoplexy conditions, diseases caused by blood hyperviscosity syndrome, venous insufficiency of the lower limbs, recovering from surgery, type 2 diabetes, obesity, hyperlipidemia and hypercholesterolemia.

6.9 Diseases of the skin cover: contact and toxic dermatitis, neurodermatitis, allergic skin reactions, fungal skin and ungual diseases.

6.10 Treatment of bronchopulmonary system: asthma, chronic bronchitis, prevention and treatment of respiratory infections (URI, ARVI, influenza).

6.11 Treatment of allergic diseases, immune deficiency, blood diseases, and others.

*) - The order of administration is indicated for the patient and for the domiciliary administration; the transdermal (no injection) administration is also possible with officially certified in Russia and Europe, Italian-made vehicles «Hydrophore» (Moscow).

**) - Without opening the metal-rubber cork of the vial, moving only the central part of the metal cap, to needle the rubber cork and draw 5/3/2 ml of the drug in the syringe, changing the needle on the syringe to the new, make an injection in the upper quarter of the buttock.

***) – There are some of functional properties in the description of products, the detailed description of the products can be found in the appropriate applications.

The main complex effects of the above preparations:

* Bactericidal: starting of specific and non-specific phagocytosis, activating microcirculation and the thymus gland, reactivation of superoxide dismutase and catalase, changes in the level of prostaglandins, the stimulation of the immune system, including increase of T- lymphocyte reproduction, optimization of the metabolism, decrease of lipid peroxidation, restoration of aerobic-anaerobic oxidation at the level of young cells (with the prevailing of aerobic oxidation), decrease of tissue puffiness;

* Stimulation of repair and regenerative processes, increasing proliferation of stem cells in the body to the level of the early ontogeny (15-25 years), activation of metabolism at the cellular level, the accumulation of ATP, increasing of fibroblast acceleration of protein and collagen synthesis, tumor of capillaries, accelerating of defect epithelialization, regeneration of organs and restoring their normal structure (liver, pancreatic gland, nephros, blood vessels, etc.);

* Stimulation of the immune response : accelerated maturation of the immune cells, increased proliferation of immune cells, including relative increase of T lymphocytes (T-killer cells), the normalization of humoral immunity, increasing reproduction of immune globulins, stem cell proliferation at the level of a young organism (magnification of 7-12 times depending on the age of the patient);



- * Analgesic and reflexogenic action: activation of the metabolism of neurons, increased levels of endorphins, increased sensitivity to pain, irritation of the nerves, the excitement of the nervous centers, the stimulation of physiological functions;

- * Anti-edema action: activation and optimization of cellular metabolism, activation of microcirculation, starting of abnormal cell apoptosis, increased phagocytosis, utilization of intercellular and intertissue liquid. More detailed explanations on the pharmacological and physiological effects of preparations are in their descriptions.

PNM effect in intracellular and extracellular processes

PNM has an influence upon the biochemical processes of oxidation inside the cell optimizing the ratio of intracellular processes of aerobic and anaerobic oxidation that are essential value in the aging processes realization, carcinogenesis, in the intracellular parasitism of pathogens and the inability to release the body as a whole from pathogens. During pathological aging an increase of triglycerides and of free fatty acids in the blood is observed. The excess of energetic substance such as fatty acids in the pericellular spatium results in their accumulation inside the cell. Intracellular balance of oxidation of two major energetic substrates such as glucose and fatty acids breaks. This means that equilibration between aerobic and anaerobic oxidation changes (the same is in aging cell or in a cell damaged by viral agent, toxin, etc.). As a result the cell switches to a very unfavourable biochemical regimen, where the effectiveness of the oxidation processes reduces.

The fatty acid coming into the cell results in an intra-cellular accumulation of triglycerides which the cell is unable to recycle. Some part of fatty acids, particularly palmitic acid, and triglycerides are substrates for the synthesis of membrane phospholipids. Palmitic acid synthesizes a sphingomyelin which has a great affinity to cholesterol. Palmitic acid, cholesterol and sphingomyelin constitute the membranous «rafts» to which the receptors are anchored. Rafts stiffening leads to receptors deactivation and the cell become «occlusive» to external signals. Thus the «devolution» of cell membranes (a kind of sclerosis or aging) is observed. «Walled-up in callous fat» cell lives as provided on its own because of losing the ability to perceive environmental signals. A damaged membrane cell ceases to adequately respond to both the signals from its own environment (chemical, hormonal, immune, etc.) as well as the impulses from the regulation centres (nerves, endocrine systems etc). Violation of the receptors function and metabolic disbalance are an initial stage of cascade homeostasis and systemic changes that eventually lead to many «diseases of aging» (hypertensive disease, ischemic heart disease, diabetes, etc.) and to immune-related disorders. When the fatty acids excessively flow into a cell they disrupt the utilization of carbohydrates and insulin receptor becomes «disconnected». Insulin resistance leads to the failure of glucose homeostasis and the content of glucose in the blood increases (the formation of diabetes mellitus) and its toxic and highly carcinogenic metabolites (glucose toxicity) also accelerate the aging process in general. Thus if the membrane receptors cease to respond to the hormone-insulin then diabetes develops, if the receptors do not interact with the receptors of cells that are responsible for immunity then the cancer may be in progress, etc.

The cells are trying to cope with the glut «bad lipids» by their reactive oxygen species direct oxidation. However, during the lipid peroxidation the accumulation of toxic and corrosive products occurs that are lipid peroxide. The content of fats in the blood and elevation of oxidative reactions producing oxygen reactive species also leads to accumulation of lipid metabolism products in the vascular walls, to the atherosclerosis development accompanied by inflammation and increased thrombogenesis, and as consequence it leads to aging processes acceleration.



PNM treatment creates the conditions for the optimization and stability of the oxidative processes balance in the cell, for its active division and operation. During such «rejuvenation» there is enough energy produced in the cell (as ATP), more effectively performed a transit of substances into the cell and removal of cellular debris, i.e. a full detoxification restored at the cellular level, the structural elements of cells are actively synthesized and «digest» the unwanted elements, including alien pathogens (viruses), a viral infection carrier state goes into elimination.

Under the influence of PNM there is an optimization of aerobic / anaerobic oxidation processes and the proper functioning of lipid peroxidation in the cell are taking place. **As a result all functions of membranes are restored:**

- **maintenance of the correct** substances gradient both inside and outside the cell, creation of a sufficient biopotential on the membrane-level. There are no prerequisites for the kinds of «aging diseases» development, related to the violation of the cells permeability (arterial hypertension, etc.)
- **proper functioning** of core micropores provides an adequate penetration of the substances involved in protection, DNA synthesis and the withdrawal of substances that threaten DNA to be damaged. Hence, mutation process of genes (oncological products) decreases
- **proper functioning** of the structural components of membranes (protein-receptor, carrierprotein, protein-biocatalysts) ensures the normal functioning of cells (the absence of reasons for the atherosclerosis development, diabetes mellitus, endocrine diseases, etc.)
- adequate recognition of damaged cells and start-up of apoptosis processes (full protection against external pathogens - antiviral effect, early detection and eradication of oncological cells)
- through impact on CD4 - receptors for HIV prevention
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Effecting in phagocytosis, as an universal method of «rejuvenation and protection» for intracellular and organism level. Due to effects on membrane structure and the creation of sufficient energy reserves in the cell there are created unique conditions **for the complete phagocytosis**. By virtue of it there is decomposition of unwanted intracellular and infectious agents to the simple substances (without the formation of residual compound products), that falls into reutilization. There is no «pigments aging» accumulating in the cells such as sediment undigested structures, the cells are functioning of full value and able to get rid of the viral load.

Influence on the stem cells with stimulation of their own stem cells production, namely cells of the spleen and red bone marrow and liver. At the tissular and organ level, this effect appears as proliferative and regenerative processes increase (e.g., proliferation of bone marrow cells increased in 13.5 times, cells of the spleen - in 4,5 times).

Effect in the immune processes. Immunotropic activity valuation suggests that the medication effective on all links of specific immunity and nonspecific resistance of an organism at all levels.

- **stimulates** the production of alpha and beta-interferons
- **mobilizes and activates** macrophages (phagocytosis enhances the absorption phase, increases the number of phagocytes, enhances the migration of macrophages), while not affecting their viability



- **limits** the formulation of inflammatory cytokines (interleukins 1,6,8, TNF – tumour necrosis factor)

- **having in possession of strong immune-activating** effect by increasing the creation of antibody-creation cells and production of antibodies to antigens of different nature. Due to interaction with antigens, allows to immune protection cells fully recognize a complete foreign agent and provide a high immune response. Similarly to it the positive results achieved in the therapy of hormonal and nervous systems disorders, caused by emotional stress with the formation of bacterial heat shock proteins, which are under the influence of PNM are successfully eliminated

- **inhibits** the viruses replication

- **increases** the organism's resistance to infections caused by viruses, bacteria or fungus disease. There is evidence for the anti-oncological effects of the medication

It reduces of HIV-infected the concentration of HIV in blood cells and plasma, activates the immune response, prevents the development of opportunistic infections. It is effective in treatment of recurrent infections mucous and skin caused by fungal agents. Stimulates antigen-specific inductive phase of immune response, at the same time increasing production of cytokines and immune cells achieving a productive phase of immune response to the elimination of antigen development. Enhances formed mechanisms of immune response. Reduces the severity of non-specific antigen reactions including anti-inflammatory reactions that are caused by TNF (by interleukin 1, and others).

The influence on the apoptosis processes – a special mechanism of self-destruction of damaged or aging cells, which is also realised during the organisms fight with the infectious diseases of the degenerative type (such as AIDS, hepatitis, etc., also oncological disease).

The result of many virus attacks and oncological mutations is inhibition of apoptosis at which point a cell attacked by virus does not die, but realises all of its pathogenic effects, and oncological cells become virtually «immortal». As a result of PNM medication effect at the cellular level (by reducing the level of transforming growth factor), the process of apoptosis functions are restored (a universal mechanism for the organism protection). Due to this to apoptosis are exposed not only abnormal cells of the vascular wall, but the abnormal cells of other tissues such as liver, thyroid gland, adrenal glands. An organism's cleanup of pathogens and oncological focuses occurs. As a result, the function of organs is restored, including the activity of enzymes strength that guarantee the cleavage of lipids, physical and chemical condition of the blood proteins is normalized, which ultimately leads to the normalization of blood cholesterol levels. Also it was found out that PNM effectively suppresses development of tumour cells colonies in cell and tissue cultures (27 stocks were investigated).

The form of medicine under analysis is intended for intramuscular injection in outpatient treatment and at home. The intramuscular way of injection is the most adequate for clinical practice, for the completeness of its bio-availability comprises 80% in comparison with the most effective intravenous injection. The substance after intramuscular injection is carried by the flow of blood and provides a general effect through the changes of intracellular metabolism. Injected intramuscularly, the medicine is absorbed quickly and completely, the time of achieving the optimal concentration in blood, extracellular and lymphatic fluid comprises from 0, 5 to 4 hours. The time of efficient influence, including the mechanisms of metabolic, immunologic and physiological changes on the cellular, tissue, organ and systemic levels which are started after the injection is from 72 hours to 30 days after a one-time injection, depending on the state of the human system and the patient's concrete pathology.



To apply as active substances preparation natural metabolites, organically «built in» metabolic ways of a bioorganism, effect of HD well studied in world medical practice allows. Currently to experts WWMA AG are known the following characteristics HD are applied in the developed techniques of treatment of various pathologies:

- Change of the object sensibility to outer and inner factors;
- The effect orientation depends on the characteristics which the object possessed before being injected the AS (active substance) HD;
- The properties of the AS SMD are changed:
 - Depending on the AS concentration in the carrier;
 - Depending on the object state at the moment of usage of AS HD;
 - The pharmacologic activity preserved, side effects disappear;
 - Non-linear “dosage-effect” dependence: the sensibility and responsiveness of biosystems to the effect of AS HD vary greatly in their force and the sign from synergism to inhibition and/or proliferation, which depends on the dosage and the solvent.

Afterword conclusion

Thus, in the event of illness that is negatively realized at the cell metabolism level and significantly accelerates the processes of aging, this medication can be used (by specially developed techniques) not only as a preventive drug but also as a highly effective therapeutic drug. The medications are effective in respect to infectious and inflammatory diseases, particularly for chronic recurrent infections, malignant neoplasms, primary and secondary immunodeficient conditions. It can be used for activation of own stem cells in cosmetology.

Summarizing all mentioned above of the PNM medication influence mechanisms at the cellular level and it is possible to define a final result of the presented pharmacological medication' effect as a shift of cells to the level of «young» functioning with all favourable results for the organism as a whole: healing, rejuvenation and prolonging life.



Medical technic equipment

• **Devices «Farmateb» and «Hydrophore» for transdermal formulation of developed drugs locally to a depth of 1 mm to 14 cm.**

Considering the system effect of devices and drugs using for regenerative process of homeostasis on parameters of fat metabolism, glucose, water-salt metabolism, target consumers of these drugs are patients suffering from the following diseases:

- Metabolic disorders - impaired fat metabolism, carbohydrate level metabolism disorder:
 - All cases of dyslipidemia (hypercholesterolemia, hypertriglyceridemia, dyslipidemia, with a predominance of LDL and HDL deficiency, etc.)
 - obesity
 - diabetes type 2 (insulin independent);
 - cardiovascular disease *:
 - atherosclerosis (cerebral atherosclerosis, atherosclerotic heart disease (coronary atherosclerosis), aortic atherosclerosis, etc., widespread atherosclerosis)
 - related to atherosclerosis conditions (arterial hypertension - various etiologies, interatrial conduction abnormality, etc.) and its extreme manifestations (heart attack, insult)
 - The combined pathology: endocrine disorders, hormonal, metabolic, and cardiovascular disease.
- *) - The term «cardiovascular disease» (CVD) brings together many different diseases: atherosclerosis, coronary heart disease, hypertension, angina, rheumatism, myocarditis, etc., as post-infarct condition.

The programs of complex treatment

Pediatrics and childbearing:

- *The program «Healthy Parents»;*
- *The program «Healthy Conception»;*
- *The program «Healthy Child - Healthy Generation»;*
- *The program «Healthy Heart since the childhood»;*
- *programs to improve the quality of life, prevention and rehabilitation process for the children with congenital diseases (CVD, neurodegenerative pathologies: CNS disorders, birth injury, cerebral palsy, etc.)*
- *The program «Childhood without antibiotics»;*



Systems of functional life of the organism:

- *prevention, treatment and rehabilitation of systemic diseases (CVD, SLM, gastrointestinal tract, SPP, NDZ, EZ, AID);*
- *prevention programs of strokes and heart attacks;*
- *The program «Healthy Heart in the middle and late ontogenesis»;*
- *therapy programs and post-stroke and post-infarct state rehabilitation;*
- *The program «Healthy stomach»;*
- *The program «Healthy joints at any age»;*
- *prevention and treatment of diseases of the musculoskeletal system: joint disease, hernia and other diseases of the spine;*
- *The program «Life without allergies»;*
- *Program «healthy liver» in any period of ontogenesis;*
- *The program «Healthy lungs»;*
- *HIV prevention, treatment and rehabilitation of urological and gynecological pathologies;*
- *potency recovery program in the middle and late ontogeny;*
- *The program «Anti-stress»;*
- *prevention and treatment of immune system;*
- *The program «Life without pain» - improving the quality of life of patients suffering from chronic diseases of degenerative type;*
- *Program «Body-fit» and «Anti-Obesity» without fitness center; Metabolic syndrome (MS) and pathology based on it:*
 - *Program «Anti-Obesity»: therapy for obesity (obesity classic - the expansion of the subcutaneous adipose tissue).*
 - *and the entire spectrum of pathologies on the basis of developing the metabolic syndrome;*
 - *Program «Body-Fit»: restore cell mass and muscle strength;*
 - *The program «Anti-diabetes»: prevention and treatment of type 2 diabetes;*
 - *MS (swelling of abdominal visceral adipose tissue) and combinations:*
 - *MS and fatty liver (steatosis), transforming to cirrhosis;*
 - *MS and pancreatitis (possibly, necrosis and pancreatic cancer);*
 - *MS and sarcopenia (loss of muscle cell mass);*
 - *MS and CVD (end - heart attack);*
 - *MS and type 2 diabetes (last stage of MS);*
 - *MS and hypertension (end - stroke);*
 - *MS and neurodegenerative diseases (Parkinson's disease and Altsgeymea, cerebral ischemia, dementia);*
 - *MS and osteoporosis;*



Age obesity:

- *atrophy of the subcutaneous adipose tissue;*
- *MS and age associated diseases (the whole spectrum of pathologies at once)*

Skin:

- *The program «Healthy Skin» - treatment of dermatitis, dermatitis and psoriasis without invasive chemical and hormonal agents;*

- *treatment of atrophic ulcers;*

Infectious and fungal diseases:

- *The program of therapy and vaccination with Candida pathology;*
- *Program of the of infectious hepatitis B, C, E, Di, etc.;*
- *Program «Anti-virus» - therapy of viral diseases of various etiologies and delivery of kits emergency assistance for the prevention and immediate elimination (1-2 days) of influenza, acute respiratory disease, SARS, cutaneous eruptions and manifestations of herpes infections without invasive drugs symptomatic medicine (antibiotics, etc.);*
- *Program «Anti-mycosis» - treatment of fungal diseases of the skin and nails;*
- *Programs of the treatment of HIV-infected and AIDS patients - until complete recovery (individually, based on the mixed pathologies);*

Neoplasms:

- *Programs of the treatment of benign skin growths: papilloma keratomas, angioma, etc., as well as diagnosis and treatment of HPV (human papilloma virus);*
- *prevention and treatment of cancer - a cure oncological diseases of different etiologies: the 2nd stage - 95%, the third stage - 65%, stage 4 - 40% (assuming no patients received chemotherapy) and increasing quality of life of patients with fatal development of cancer pathology;*
- *support programs among patients with cancer undergoing chemotherapy and radiotherapy;*
- *Programs of melanoma treatment*

Gerontology, geroprophylaxis:

- *rejuvenation program of functional systems of the body;*
- *The program «90 years without osteoporosis»;*
- *prevention and therapy of age-related diseases: sarcopenia, osteoporosis, neurodegenerative disease, etc.;*
- *Program «Anti-Aging»: a set of measures to conserve the optimum balance between subcutaneous and visceral adipose tissue.;*
- *geriatric program «One hundred years without age-related pathologies»;*
- *geroprophylaxis program;*
- *healthy longevity program (non-pathological reaching the age of 80-90 years with a high quality of life)*



Created on the basis of: medical equipment, medical drugs and methods of its use, the programs of comprehensive treatment of a variety system and age-associated diseases were developed: diseases of the cardiovascular system, including developing against the background of the metabolic syndrome (hypertension, hyperlipidemia, diabetes type 2, obesity, etc.), gastrointestinal tract and the musculoskeletal system diseases, urological and gynecological diseases, infectious diseases of bacterial, viral and fungal etiology, including skin, skin and venereal diseases and Mycosis. The special focus for Companies group is geriatrics: neurodegenerative disease - senile and vascular dementia, Alzheimer's and Parkinson's (tremor), sarcopenia, osteoporosis and others.

Summary

Long-standing practice of the majority described devices and medical drugs, clinical research and in-depth analysis of its using has shown that comprehensive prevention programs, treatment and rehabilitation can replace more than 3000 drugs used in the prevention and treatment of the entire spectrum of diseases in both adults and children of all age groups. The use of program approach to the treatment the body with all complex pathologies provides patient as prevention of many diseases of age and cure most of them, including considered to be chronic and incurable, without having any negative side effects. The effectiveness of different diseases treatment reaches an average of 80 to 95%.

In «neglected cases», including cancer 3rd and 4th stages, the rate of successful treatment drops to 60%, but it provides a systematic improvement of body functioning and improving the quality of life of critically diseased patients.

The entire complex of programs for its therapeutic properties provides patient recovery and healing, excludes diseases transferring to a latent form and invalidization in late ontogeny condition, replaces a dozen visits to a medical institutions, financial assets economy but leaving it necessary to periodically visit the independent (but highly professional and quality) diagnostic laboratories and research centers on Holding doctors profile recommends.



CURRENT RESEARCH WORK of WWMA AG

Today, in the field of immunology, pulmonology, cardiovascular system diseases, infectious diseases of bacterial, viral and fungal etiology, musculoskeletal system diseases and the type of degenerative diseases (cancer and neurological) in the active phase runs on the following fundamental and applied scientific work of scientists team of WWMA AG and its subsidiaries:

- 1. Develop a mechanism for regulation of mitochondrial activity and its use in solving the problems of infectious diseases and longevity of bioorganisms;*
- 2. Develop a mechanism for the formation and launch of specific autoimmune vaccine (endogenous) in the human body against bacterial infections and viral etiology;*
- 3. The study of the mechanism of maternal inheritance extrachromosomal and use it for solving problems of infectious and non-infectious diseases laboratory and farm (pigs, cattle) animal - within the framework on modeling of such processes in human body;*
- 4. Simulation in laboratory and farm animals, the process of active Tkiller formation of immune system in the neonatal period and its prolongation in the postnatal period in the phenotype;*
- 5. Development of the drug, fixing a protective immune organs of low molecular weight components of microorganisms and methods of its use for the comprehensive protection of human infectious diseases;*
- 6. Development of a method and preparations to activate specific complete phagocytes viruses and other intracellular pathogens to solve the problems of elimination of infections including viral etiology influenza, herpes, chlamydiosis;*
- 7. The theoretical substantiation and obtaining experimental results in the development of specific antibodies shipping procedures inside the ontogeny cell (innovation), to develop a comprehensive approach to the prevention and treatment of cancer: the organization of industrial production of drugs for the effective treatment of cancer of various etiologies, including 3rd and 4th stages (before the process of ascites);*
- 8. The development of drugs and techniques recovery of human reproductive activity - innovative approaches to solve the problem of infertility;*
- 9. Development of innovative approaches for the treatment of human tuberculosis by activating specific complete phagocytosis of mycobacteria and brucellas by stages: 1st identifying the basic protein fractions against Mycobacterium TB 2nd stage of identifying the type of phagocytosis on laboratory animals, 3rd phase of process modeling elimination of mycobacteria on farm animals - cattle, development on the basis of these methods-line drugs for the treatment of human tuberculosis to complete elimination of pathogen from diseased cells of the body;*
- 10. Development of a method of epigenetic regulation of intracellular processes of methylation and redox potential - the main element of the system of maintaining intracellular homeostasis.*



The work includes: development of the metabolic scheme of intracellular methylation, its pairing with redox potential, the energy transformation in the mitochondria (the regulation of tricarboxylic acid cycle and oxidative phosphorylation) and the synthesis of lipid products: fatty acids, triglycerides, phospholipids - phosphatidylcholine and sphingomyelin, cholesterol. A single mechanism of cell energy flows regulation was identified, DNA replication and proliferation, wherein the low molecular weight compounds involved. The modulation of concentration of low molecular weight compounds allows epigenetic effects on genes expression encoding protein regulators. Full mechanism description of epigenetic regulation has created a theoretical and experimental basis for the development of a new generation of pharmaceuticals, allowing the system to ensure maintenance of homeostasis.

Based on this fundamental theoretical foundation the lines of new generation medical drugs development is being run for prevention and treatment of systemic and age-associated diseases.

11. The duration of comprehensive exploring issues of developing metabolic syndrome - main reason of age-related diseases and senescence:

- etiology of metabolic syndrome;*
- diagnosis of the metabolic syndrome;*
- regional distribution of adipose tissue and the metabolic syndrome in late ontogeny;*
- age-related metabolic changes in the pathogenesis of late ontogeny;*
- age-related diseases - a consequence of adaptation to changes in the internal environment;*
- metabolic syndrome, obesity, fatty hepatosis;*
- free fatty acid - a marker of age-related changes in homeostasis and aging;*
- age-related changes in energy metabolism, later ontogeny disease and aging;*
- metabolic routes at the cellular level and markers of age-related pathologies and aging;*

12. The duration developing on the basis of above said researching works of new generation, geroprotectors for the prevention and treatment aging diseases to promote the healthy longevity.

The developed new types of medical drugs are highly pharmacological and physiological efficiency, which returns man homeostasis to the optimum level in any period of ontogenesis, taking into consideration patient's phenotypic changes while treatment. Due to small molecules and ultra-low doses of active substances - natural body metabolites and vitamins in an isotonic solution, these medical drugs do not have any drug side effect to human body.



INNOVATION, completeness and TEST DEVELOPMENT.

1.1 The **goal** of developments is **Innovation**: «The principled creation of a new generation of drugs and techniques in all sectors of bio-organisms: plants, animals and humans the main reason of researching the basic mechanisms of metabolic and functional activity of the cell bio-organisms on molecular level, that allowed to **make a number of important and fundamental discoveries** that were used in creation of drugs and methods in the following areas:

- **human medicine** - for early diagnosis and treatment of systemic diseases, as age-related diseases affecting the lifespan; prevention of infectious and aging diseases, post-operative rehabilitation;

- **pharmacology** – the principled creation of a new drugs based on natural metabolites, including endogenous to the human body in origin of extra small and nano-dose using as dosing vehicle -physiological saline (NaCl)

- for parenteral injections, and phosphatidylcholine - in the shape of nano-capsules carrier of the active substance (with a size of 10 nm);

1.2 The base discoveries in pharmacological industry of Company

WWMA AG are:

- development of **methods and diagnostic tools for determination of biochemical and pharmacological activity of natural and synthetic substances in ultra small and nano-doses**. Given technique allows accurately identify the « dome of pharmacological activity» of a substance, the number activity zones of the substance on the scale of micro and nano-doses and orientation of the pharmacological action of the drug (proliferation or inhibition), its «lower and upper border» in the short term (up to 30 days), which provides creation of a new drug for specified goals in a significantly shorter time (up to one year);

- development of technologies for creating bio-carrier on the base of using endogenous substances in excess of small sizes (bio-nanocapsule up to 5 nm) with biochemical elements inside to direct transportation through the cell membrane.

1.3 The created line of drugs **have not existed until now pharmacological and physiological efficiency, restoring homeostasis of all bio-organism systems to the optimum level at any stage of ontogeny with the phenotypic changes** while treatment.

1.4 For the first time ever in the world the pharmacological drugs of new generation that do not have any adverse side effects on the bioorganisms were created.

1.5 **High functional (pharmacological) efficiency combines with high economical (cost-effectiveness of production and the use of drugs)** due to:

- the use of nanotechnology to produce a concentrate and standard equipment for filling ampoules;

- due to the classical method of using - parenteral injections (intramuscularly or transdermal routes) in hospitals, outpatients and home treatment.



1.6 The completeness and the level of development in all three areas: medicine, veterinary science, technology of growing farm animals, birds, fish and agriculture - are divided into three groups:

- **the know-how in the field of fundamental knowledge (theories) the origin and development of biochemical reactions**, which provide metabolic processes at the cellular level and **functional, physiological processes in humans and animals**: immune and phagocytic, the effect of mitochondrial activity, and many others: the part – is already confirmed and registered, and part – is going through laboratory testing of drug samples, procedures and accepted hypotheses;
- **theory, drugs and procedures that are protected by copyrights and patents** that have passed the stage of laboratory and industrial tests, and fully proven;
- drugs and procedures that have passed full or partial clinical research are currently being used in medicine - as a part of clinical trials on volunteers for the prevention and treatment of various diseases, in veterinary medicine - for the prevention and treatment of most diseases with agriculture and domestic animals (cattle, pigs , sheep, etc.) and birds.

1.7 The completeness and level of procedures of created drugs and methods:

– all created pharmacological drugs: two lines in the medicine field - forty drugs, ten veterinary drugs, and twelve drugs for use in agriculture (cattle breeding, poultry farming, fish breeding and agronomy) – **have passed the full range of necessary laboratory, pre-clinical and clinical (in-vitro and in-vivo) testing, a number of drugs are being used on human volunteers** as a treatment of incurable diseases (hepatitis B and C, etc.) on experimental terms and obtaining the effects of restoring individual systems and bio-organisms in general, and - after appropriate testing were **registered and being used industrially – the drugs for the rehabilitation and treatment of farm animals** (cattle and pigs) are in the process of experimental and industrial test preparations to provide additional weight gain of animals, chickens and fish, the drugs for grains treatment (rice, wheat) to increase grain yield.

2. IP List - the know-how (theories, hypotheses and articles with discoveries) and patents in the area of human medicine in a dozen countries (Europe, Japan, China, Singapore, UAE, Brazil, USA, Russia, CIS).

3. The market - in each specified area - there is a separate segment of market where created drugs and procedures will be highly sought immediately after product launch. The current situation of economic crisis strengthening in most countries, significantly increases the chances of a quick introduction to the market the relevant services - procedures and products, ten times increases the efficiency and effectiveness of impact, hundred times exceeding economic growth (which means the possibility of saving money while increasing the efficiency but not vice versa, as it happens commonly nowadays) of the drugs and procedures in these areas of economy in any country.

- In the medicine area:

- **a segment of incurable, invasively curable, ineffectively curable** or transferred to the latent and chronic infections continues;

- currently the most important segment in meaning and scope is a widespread - covering up to **60-70% of the adult population (35% of the world population over 40 years) the spread of cardiovascular system diseases and diseases associated with metabolic syndrome**, including different forms of obesity that have found an effective solution by using the developed procedures;.



- programs of **recovering reproductive human qualities** with ensuring the birth of offspring with physical and mental IQ in the range 9-10;

- a significant segment of the global medical market that is lack of adequate offers to the existing demand, are **infectious fungal diseases**: chlamydia, aspergillus, mucorosis, candidiasis, and others. From 20 to 40% of the adult population in developed (the level of morbidity is higher) and developing countries is annually suffering from these disorders, while the products that were created by us provide both therapeutic and prophylactic effects and can be used in the existing infrastructure medical centers;

- the special segment of the market is **the market of early detection of cancer diseases**. In spite of existing offers in this segment, they relate to only a few forms of cancer and the best case account for 50% efficiency, while proposed methodology allows to determine any availability of cancer diseases at an early stage, non-invasive for patients and gives the answer «yes or no» within 24 hours from 90% probability. The volume just of this market in Europe - 125 million diagnostic cases per year (2008) while treatment with complicated, expensive and inefficient methods of diagnosis - the demand is covered by only 40%. In view of the tendency of twice growing the number of cancer disease in over the next 5 years and the WHO recommendations for the population - to take the diagnosis at age over 40 years twice a year - the growth rate of this market segment will continue to significantly exceed than the appropriated offers.

- the following development segment of the health care market is **the treatment the patients with acquired immunodeficiency syndrome (AIDS)** - today the market offers only the drugs and procedures to slow down the rate of lesions development (viral load) of the patient. Programs offered by WWMA AG that has passed all phases of clinical research, provides the reduction of viral load at any stage of the disease to the values at the time of being intentioned, it means the transferring of HIV patients in the HIV status. This project provides a place for all international and national programs against AIDS.

- currently the most dynamic medical market of services in the following fields: **geroprophylaxis, gerontology and rejuvenation** - the volume of the market is much greater than the available offers, and most of the offers do not suit any customer requirements, neither the declared goals of the proposed methods; market sector includes both prevention and treatment of chronic and age-related diseases (dementia, Parkinson's, Alzheimer dementia), and special programs for the regeneration of tissues and organs, rejuvenation of the whole body, which may be the object of sale to existing profile centers and / or to be the subject of creating an international network gerontoprophylaxis VIP centers and Clubs of health longevity, exclusively owning developed on the basis of established lines geroprotectors integration programs for high class patients of different countries.

Example: - the actual **pandemic of infectious forms of hepatitis «B» and «C»** incurable to the extent of 100% elimination of the disease at the present time, only in China - 50 million (officially) to 100 million (effective) people are carriers these diseases, no less - in India and Sri Lanka, the diseases wide spread occurrence in Latin America, the CIS and even Europe - **the volume and depth of this segment of the global healthcare market is much more than the above amounts and the procedures and drugs of WWMA Ag successfully completed two phase of clinical researching and being used in practice.**



- **The sector of prospective research projects** of the company and its subsidiaries also covers a lot of the resolutions that set out in the world, and in particular provides for the problems solution related to the category of innovative technologies of future generations. A separate market segment in the company presented by **a special software product (PP) developments, for the purposes of medical off and on-line patients supporting with corporate and individual use.** The particularity of the PP is that after being embedded in number of countries, it will allow to form (by input parameters and analytical data) healthy mode of life. **The ultimate goal of the PP development is creation of a organism model, a kind of virtual «twin» of man that allows to prevent the occurrence of diseases or to diagnose them in a very early stages, and also accompany treatment and rehabilitation processes - which will promote the formation of a new - physiologically and psychologically - type of a person.** Creating such a system is not possible without a number of fundamental discoveries that made by scientists of WWMA AG in the areas described above and developed on basis of diagnostic, preventive and therapeutic, procedures.

The total market segments and solvable problems indicate the actual creation of a new industry – the industry of functional life and bio-organisms at the cellular level regulation, which allows the control of individual systems and the organism itself in order to achieve an optimal state of homeostasis of bio-organisms in any period of ontogenesis and solve nowadays urgent problems of life support - most effective, with a functional and economic points of view, that set up in all countries. These conclusions are objective and confirm unconditional demand on Company's developments, services said above, in the world market.



LABORATORY OF DIAGNOSTICS for SYSTEMIC DISEASES

TERMINOLOGY: *systemic diseases - cardiovascular diseases; neurodegenerative type ALS, Alzheimer, MS, Parkinson tremor; metabolic syndrome; type 2 diabetes; obesity; oncology.*

Diagnostic laboratories / methods:

1. EEDOD (early express diagnostic of oncology diseases (any etiology on cell's level)) - laboratory performs early express diagnosis of cancer of any etiology on cellular level. The patents on the method of diagnosis: Knowhow – diagnosticum, designed and produced by WWMA AG, allowing to analyze and signal about the presence of or predisposition to oncology disease (with its expected manifestation within 1-2 years after diagnosis) in 24 hours.

2. METABOLIC diagnostics - laboratory diagnoses systemic and age-related diseases on the metabolic profile of patients based on the identified markers of the state of energy homeostasis. Methods and markers have not yet been patented. Know-how - specific markers, separated by the scientists of WWMA AG.

3. The diagnosis of homeostasis - is based on the premise that all systemic disorders have common characteristics that describe the change in the parameters of the stationary state of the body - homeostasis and biochemical, physiological and mathematical model of homeostasis, developed by our research team. This model is a model of management of REDOX system of body, which optimizes homeostasis by controlling the following processes:

- apoptosis; • proliferation; • phagocytosis; • synthesis of catecholamines;
- mitochondrial energy mechanism;
- and other functional systems of life at the cellular level, thus providing management of information and energy «space» of the body.

This model describes:

- redox system of a body;
- Krebs cycle - mitochondrial energy mechanism;
- pentose phosphate cycle (recharge of protic flows);
- urea cycle (amino group - additional controller);
- cycle of one-carbon compounds (methylation of DNA, RNA, epigenetics);

The model describes the Redox system as coherent non-hierarchical system and provides the following definition of the body pathology - pathology is an adaptive response of the body (meaning - redox system) to change of the settings of coherence cycles that provide functional activity of human body, which is expressed in the destruction of a single rhythm of cycles and their nonperiodic oscillations.

Treatment programs are developed as per identified diagnostic data basing on the use of two methods of management of redox-system of the body:

- using a separated key (archaic) metabolites, dissolved in ultra-low doses in saline solution, administered as per program under different states of the body;
- small smart molecules, synthesized by WWMA AG, allowing to control the Redox system by proton «blows/impacts».

Know-how of WWMA AG: our team discovered general and specific markers that characterize the state of the human body, based on studies of lipid and carbohydrate metabolism, hormonal status, catecholamine status, oxidative stress, systemic inflammation, nature plaques on the carotid arteries and in the pool of lower limbs.



Patterns of these markers allow us **to determine the stage of pathology and predict its development**, taking into account age and gender differences.

Individual treatment and prevention programs are developed on the basis of all three diagnostics.

Objectives: To improve the quality of life of patients suffering from degenerative or chronic systemic diseases, therapy of patients with these diseases depending on the stage and condition of the body (the degree of accumulated disorders in it, from a particular disease or mixed), slow down and / or stop the development of systemic disorders, as well as prevent their transition into a pathological phase, thus ensuring a slow non-pathological aging and healthy longevity.

The results of the diagnostic evaluation of patients:

The results of a comprehensive survey allow us to detect the type of disorder in energy homeostasis and predict the course of system processes (pathological, evolutive).

According to the study we provide recommendations for the prevention of sequela and reduction in the rate of development of the system process with respect to evolutive and pathological processes.

Implementation of the recommendations for the prevention of acceleration of system evolutive processes is done by appropriate program of therapy based on drugs and methods of WWMA AG.

Overview of developed products scientific base

Having set the objective of developing medical technology, medicines and techniques for securing effective human life of the new generation, we investigated basic mechanisms of metabolic processes and functional activities of bioorganism cells at the atomic and molecular level. Additionally, we conducted retrospective analysis of metabolism evolution from the moment of its appearance on earth 3.6 billion years ago. This allowed us to make a number of important and fundamental discoveries:

- The methods and diagnostic tools to identify areas of biochemical and pharmacological activities of natural and synthetic substances in micro and nanodoses that provided an opportunity to create a new generation of pharmaceuticals;
- The technology of small molecules of a given size and «content», and methods of their application for therapeutic purposes;
- The theory and studies of mechanisms of epigenetic regulation of homeostasis in functional cells of multicellular organism systems and small molecules;
- Working model of evolution of metabolic processes in living organisms, from the moment of transition from the stage of chemical evolution to the biochemical one: emergence of initial forms of living matter - archaea, methanotrophic and methanogenic bacteria, on the basis of which we created diagram of metabolic processes that maintain energy and biochemical homeostasis of all functional systems of somatic cells of bioorganism;
- Generalized scheme of metabolism of human somatic cells, which allows to identify key metabolic pathways and metabolites, markers of pathologies and aging, which had taken its evolutionary place and objectively characterized the state of biochemical processes, tissue, organs, and their systems at the organism level, playing key role in these processes .
- Lipid theory of aging of the human body;
- New generation of pharmacological preparations having both generalized (immunomodulating, regenerating) and specific (preventing and curative) effects on the body.



THE EVOLUTION OF UNICARBON COMPOUND TRANSITION SYSTEM FROM AN ABIOTIC PATHWAY FORMED 3,6 BLN YRS AGO TO A HUMAN CELL. FUNDAMENTAL BASIS OF METABOLISM.

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EVOLUTION of CELL-METABOLISM

The large number of biological structures acquires a complete spatial form with participation of methyl groups. Constant concentrations of methyl and amino groups are strictly maintained in the biological systems to make it functional and stable.

The biosystem functions within the framework of oxidation-reduction cycle in the field polarized by phosphorus. The biological system is reserved in the polarized field and constructs itself as an autonomous redox cycle. **Autonomous biosystem is built into the common planet space in global oxidation-reduction cycle.**

Three redox cycles are united in global "hydrogen-oxygen" cycle

Redox cycle of nitrogen: ammonia - nitrogen oxides - molecular nitrogen - ammonia.

Redox cycle of sulfur: hydrogen sulphide - sulfur oxides - molecular sulfur - hydrogen sulphide.

Redox cycle of carbon: methane - methanol - formaldehyde - formic acid - carbon dioxide - methane.

The last is a beginning of life in the form of methanogenic and methanotrophic archaea.

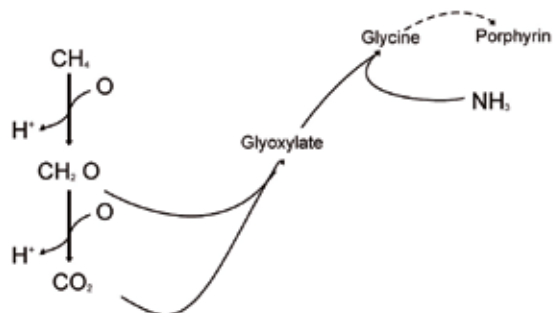


Fig. 2. In biosystem evolution there takes place a transition from unicarbon to two-carbon compounds (acetate synthesis). Formation of the reductive tricarboxylic acid cycle. Divergence of formic aldehyde and carbon dioxide ways of utilization.

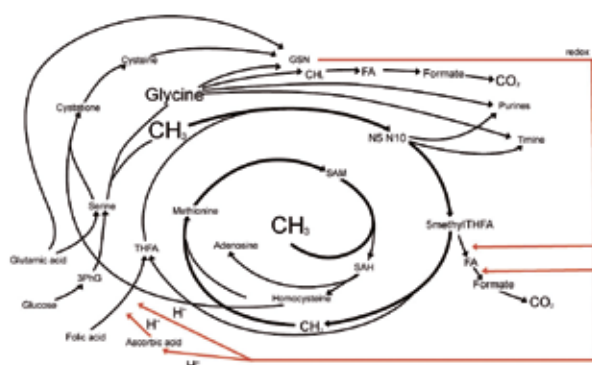


Fig. 1. Methanotrophic metabolic pathway for unicarbon compounds. Glycine and glyoxalase formation. Porphyrine synthesis out of glycine. Metallic porphyrins - the possibility for molecular oxygen, protons and electrons translocations, photosynthesis and respiration.

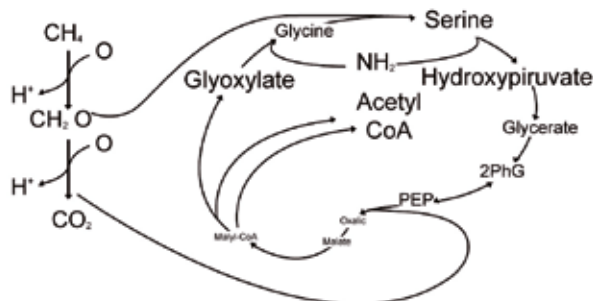


Fig. 3. System of unicarbon compounds transportation in a human somatic cell. "Glycine>serine" inversion produce a methyl group for methylation and oxydation. Oxydation goes on by the way "methyl group-methanol-formic aldehyde-formic acid-carbon dioxide" and is an archaic source of protons and endogenous carbon dioxide molecule. Formic aldehyde is a modulator of proton homeostasis in cell.

The stationarity of the protons and electrons streams managed by the system of the methyl groups transfer and their oxydation is the fundamental bases of homeostasis in the biosystems.



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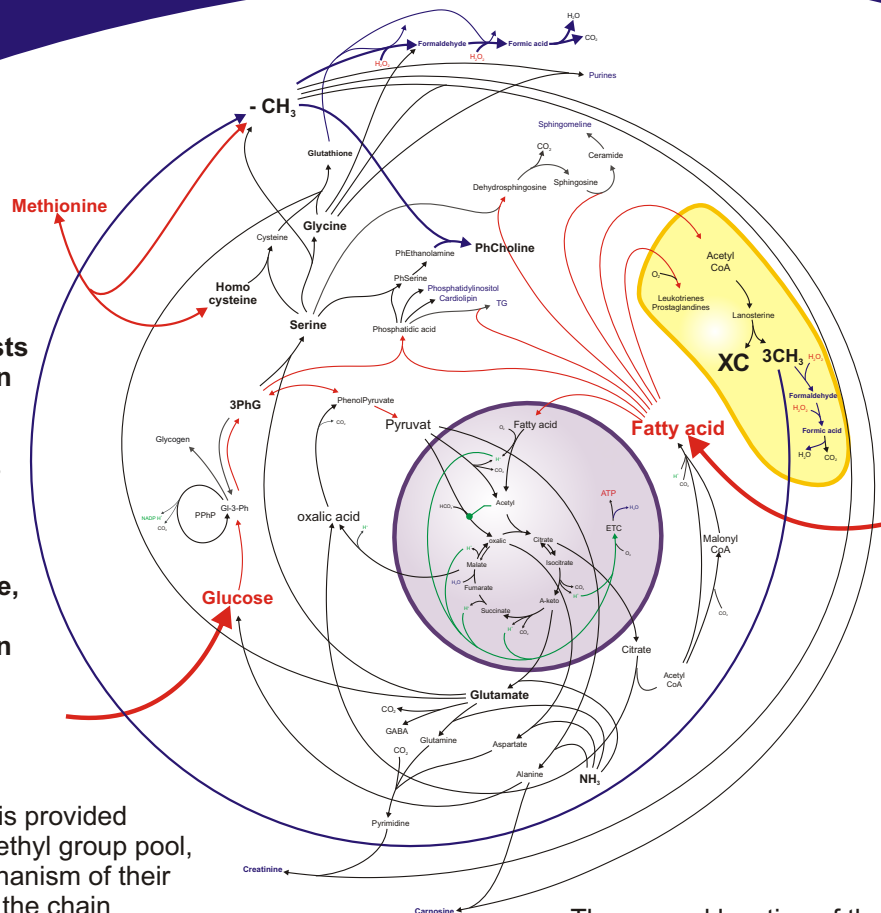




Generalised scheme of human cell metabolism

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The metabolic scheme designed by WWMA specialists allows to distinguish the main junctions of metabolic ways, by which the connection of electronic and protonic flows is performed and main physiological functions are controlled, such as energy consumption and expenditure, synthesis of biological macromolecules, proliferation and apoptosis.



Functioning of cell metabolism is provided by maintenance of constant methyl group pool, which is controlled by the mechanism of their gradual oxidation according to the chain “developed” by methanotrophic bacteria: “methyl group – methanol – formaldehyde – formate”.

As for eukaryotes, the oxidation of methyl groups is performed by enzymes which are involved into the system of antioxidant defense, or detoxication, located in microsomes. This way of oxidation is so strictly regulated by eukaryotes that the excess of formaldehyde and formate causes immediate death, i.e. these compounds are virulent poisons. These metabolites perform excessively important functions, therefore their homeostasis is strictly regulated. Actually, this is the key metabolic and functional junction, the destruction of which causes immediate death.

The second location of the “methyl group – methanol – formaldehyde – formate” way is in peroxisomes, where monocarbonic compounds are connected with catalase. This way is controlled by factors which regulate the activity and expression of catalase.





FUNDAMENTAL INVESTIGATION BY WWMA AG

In the world there exist more than 300 theories of Aging: genetic program, accumulation of oxidized rubbish, reproduction termination, calorie restriction. None of them answer the main question of Humanity:

Answer is found:

Human organism is a thermodynamic system in which glucose energy is transformed into the fatty acids energy. Glucose is a proton donor while fatty acids are their acceptor.

The level of protons, or pH, is the basic stationary indicator. We have discovered the existence of a proton flows regulation system in cell.

Free fatty acid homeostasis changes in ontogenesis

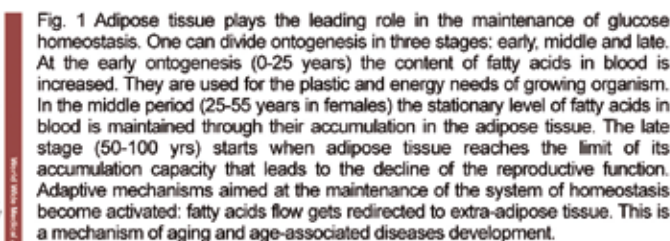


Fig 2. Adaptive mechanisms to the extracellular fatty acid abundance: membrane components synthesis, fat accumulation, apoptosis, lipid oxidation (oxidative stress).

Consequence of elevated free fatty acid blood level

- Hormone resistance
- Changes in signal transduction
- Fat accumulation in nonadipose tissues
- Apoptosis
- Oxidative stress
- Cancer

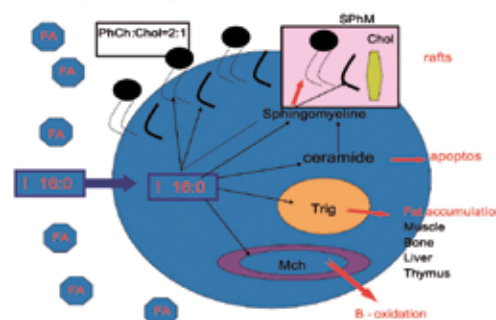
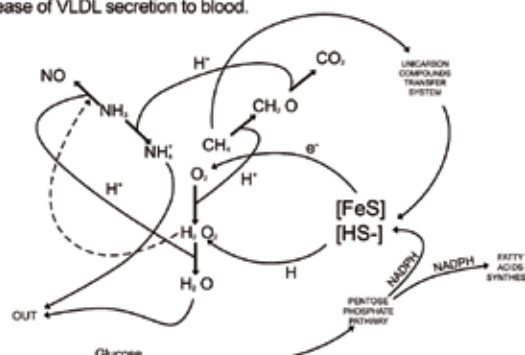


Fig. 3. Adipose regeneration of muscular tissue causes apoptosis and sarcopenia, while that of bone tissue results in senile osteoporosis. Excess of fatty acids inflow to liver provokes the adipose regeneration of liver (steatosis) and the increase of VLDL secretion to blood.

Fig. 4. The accumulation of triglycerides is the irreversible thermodynamic process which leads to the termination of the system functioning. Redox cycle is the basis of homeostasis in the biosystem and its functioning in stationarity.



Endogenous synthesis of fatty acids completes the rotation of the redox cycle - Life goes away.
Fatty acid production and accumulation in the human body is evolutionary determined process.
During life the body is converted into a fatty crystal.



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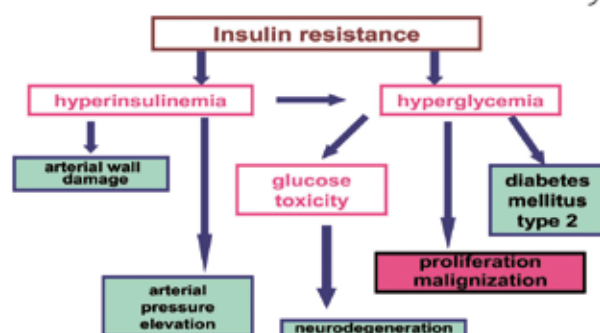
Age-associated diseases

Fundamental investigation by WWMA AG

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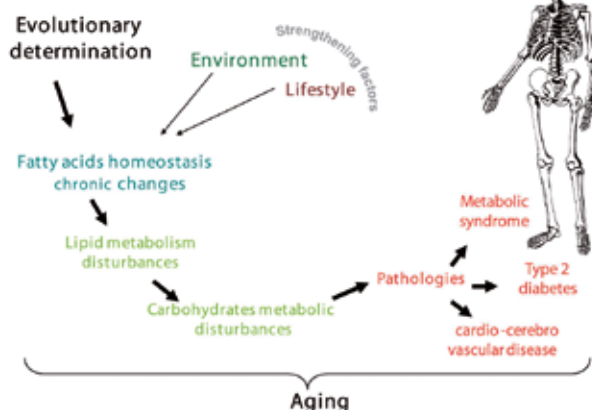
Human organism is a thermodynamic system converting glucose energy into a completely reduced compound - fatty acid.

Adaptation during aging is aimed to sustain the stationarity of fatty acid distribution by fat accumulation in extra-adiposes tissues. It leads to the insulin resistance and desadaptation.



Insulin resistance is a reason of arterial wall damage, neurodegeneration, malignization (cancer), diabetes mellitus type 2.

Lipid Theory of Human Aging



The PNM action is aimed to prohibit the intracellular triglyceride synthesis and activate mitochondrion respiration that taken together elevate insulin receptor sensitivity.

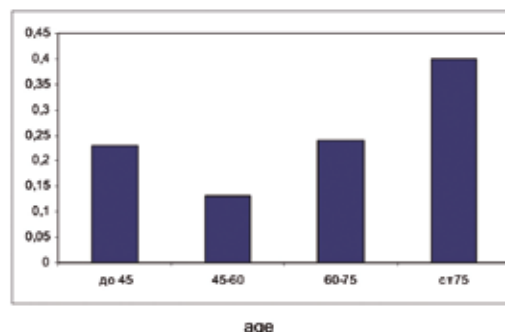
Age-associated diseases are systemic pathologies

- Cardiovascular
- Neurodegenerative
- Atherosclerosis
- Diabetes mellitus 2 type
- Osteoporosis
- Cancer
- Pernicious anemia

The common mechanism of aging

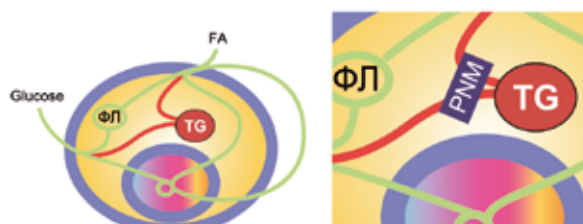
Fatty acid level in blood is elevating through the last period of life. Fatty acids level is a marker of the systemic homeostasis failure and age.

The age distribution of patients with FFA > 30 mg / dl



Lipidic Theory of Human Aging persuades that the Aging can be regulated.

PNM as a modulator of intracellular red-ox balance



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MEDICINES AND PROGRAMMES DEVELOPED BY WWMA AG

FUNDAMENTAL INVESTIGATION BY WWMA AG

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Properties of the preparations based on the natural metabolite (PNM)



Anti ageing, anti stress and healthy generation programs developed by WWMA AG

The programme includes 4 stages:

- diagnostics of patient's current health state and assessment of patient's actual biological age;
- treatment of detected pathologies;
- "anti-aging" programme implementation;
- maintenance of the reached state of organism's rejuvenation during the longest possible period of time

Results achieved:

- termination of organism's chronic pathological states;
- restoration of lost functions;
- prophylaxis of systemic disorders of late ontogenesis;
- deceleration of aging process;
- rejuvenation ascertained through the control of biological age markers such as: subcutaneous adipose tissue atrophy, decrease in the level of sex hormones, increase of the homocysteine level in blood, bone tissue density decrease, left ventricle hypertrophy, appearance of pigment spots on skin – lipofuscin accumulation, collagen glycosylation – skin flabbiness, wrinkles, grey hair, accumulation of microelements in the hair and nails, increase of cobalt content in urine, ALT/AST ratio.

MEDICINES AND PROGRAMMES DEVELOPED BY WWMA AG

XR-GERONT - a new gerontology medicine (geroprotector) and method of its application.

XR-REP - medicine for recovery of reproductive function and method of its use against infertility problems.

XR-AV - medicine for activation of specific complete phagocytosis of viruses and other intracellular pathogens in order to eliminate infections of viral etiology, including flu, herpes and Chlamydia.

XR-AH - medicine for complete elimination of B and C viral hepatitis and method of its application.

XR-APP - medicine for treatment of neurodegenerative diseases typical for late ontogenesis, such as Alzheimer's and Parkinson's diseases and its application method.

GP, GP/PQ - medicines and methods of their application for metabolism normalization in human organs and tissues. These medicines allow:

- to prevent metabolic disorders and ensure cell metabolism recovery;
- to prevent and treat disorders in musculoskeletal system.

Medicines and methods of their application for medical and preventive treatment of infections of mycotic etiology.

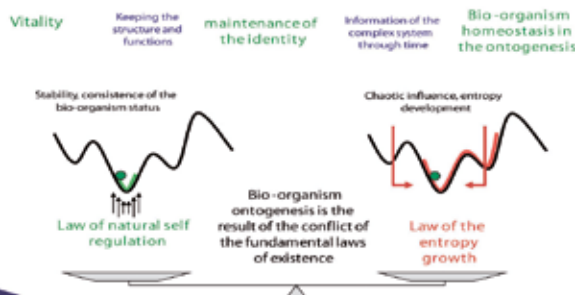
Medicines and methods of their application for AIDS treatment

A line of medicines and methods of their application for treatment of pathologies of different etiology (bacterial, viral, mycotic) for oral administration and cutaneous application on the base of metastable solution of different microelements: H, O, Fe, Mg etc. in the ionized (liquid) form.

A line of cosmetic medicines for anti-aging treatment.

Special program for anti-aging treatment: somnology, readjustment of bioenergetics etc..

Ageing-general decrease of vitality with advancing of age



PNM effect - infertility



19 family couples (from 26 to 43 years old) had registered infertility during 5-20 years. It was defined with various abnormalities of ovulation cycle, endometriosis, cysts of ovaries and other female pathologies, and cases of male and joint infertility as well. Use of different well-known therapeutic methods did not result in getting aim. After cure with PNM a positive effect was achieved by 16 family couples – 17 healthy children were born with 9-10 mark of health by Apgar scale. Children born by those parents are developing in normal way. At present time the oldest of them is 9 years old.



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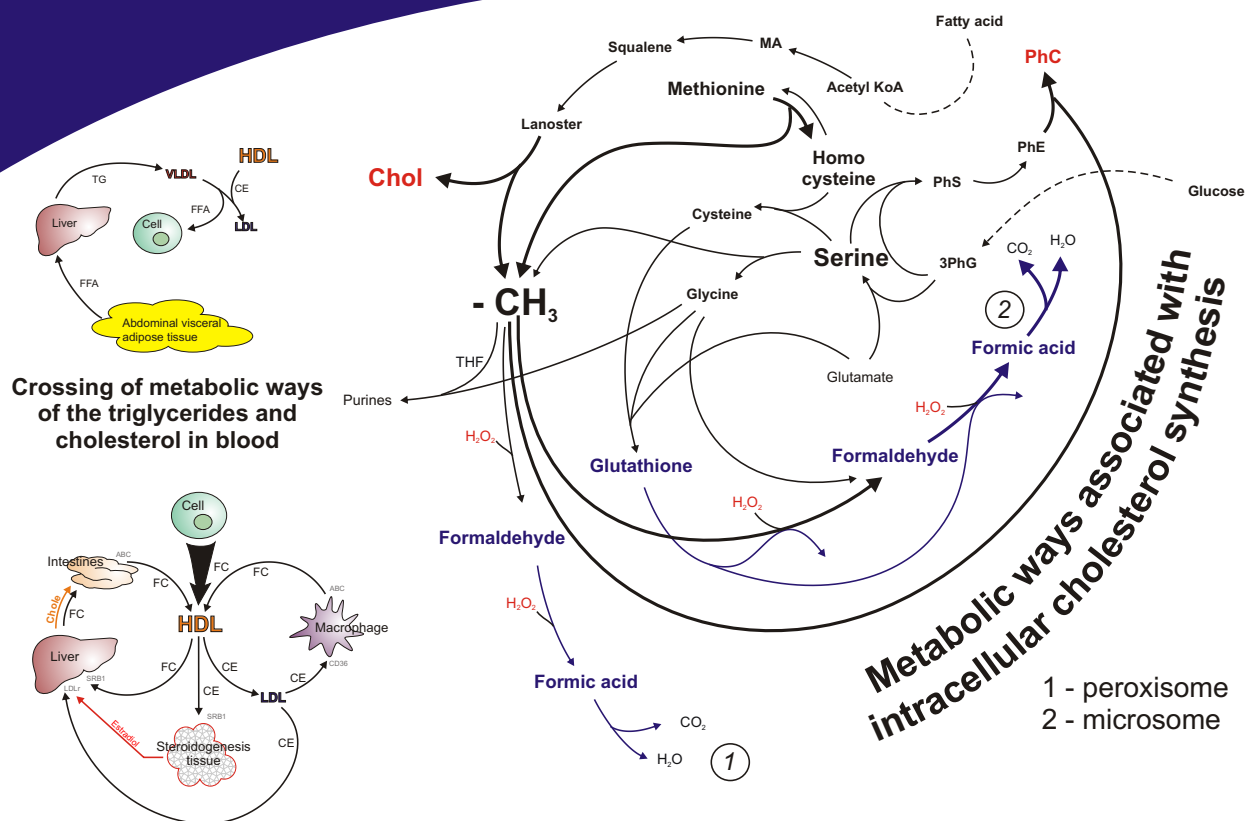
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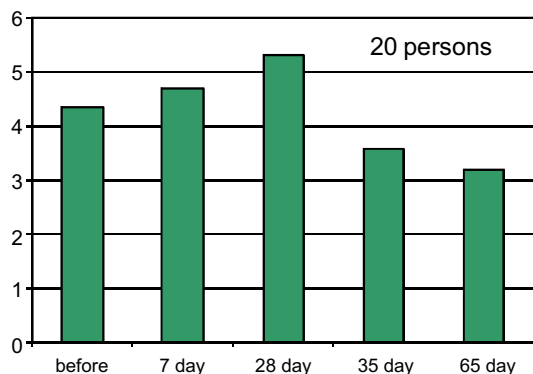
Cholesterol lowering effect of formic aldehyde injections in super-low doses

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Cholesterol recirculation in blood

FA action on cholesterol content in the blood (Mmoles/l)



FA action on carotid arteries (1 year period)



	2008.10	2009.04	2010.03
common carotid artery (right side)	30%	27%	24%
common carotid artery (left side)	43%	38%	28%
internal carotid artery (left proximal part)	48%	44%	37%
artery wall differentiation	absent	absent	restored
AP systolic	135-145	120	120
diastolic	90-100	80	80



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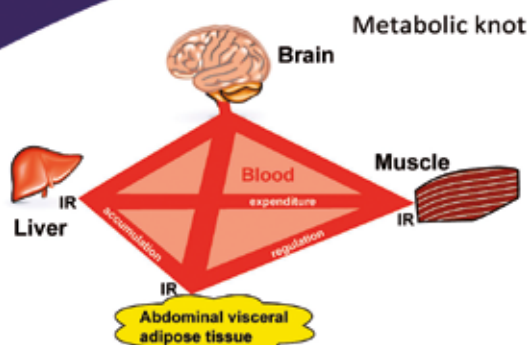


Mechanism of neurodegeneration

FUNDAMENTAL RESEARCH from WWMA AG

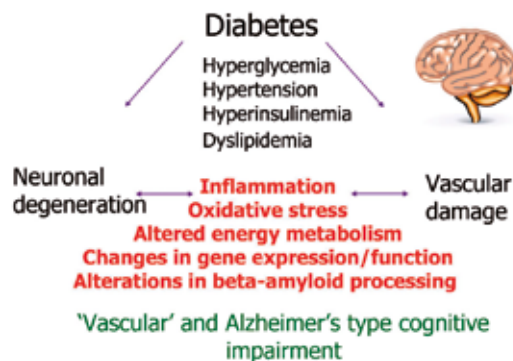
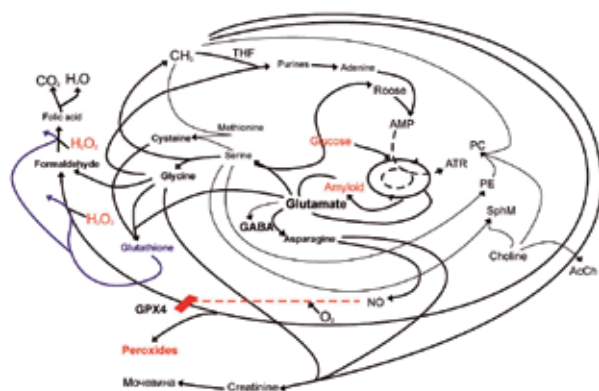
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We have designed a conception of "the metabolic knot" as an octahedron

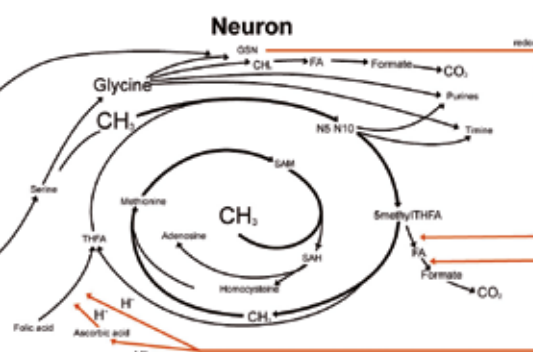


In the tops of the triangular foundation there are liver, muscle and abdominal visceral adipose tissue where insulin receptors are expressed. Thus a glucose stationarity is achieved and sustained/ There connections: "brain-skeletal muscle" (glucose delivery), "brain-liver" (cholesterol delivery), "brain-adipose tissue" (fatty acid delivery).

Brain has the same unicarbon compound transition system as all human cells but it is divided between astrocytes and neurons. Glucose is a donor of protons and neuromediators (glycine and glutamic acid).

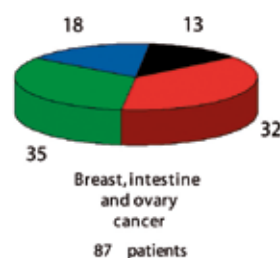


Aged associated neurodegeneration is a senile dementia. Etiology factors are glucose stationarity impairment and carotid arteries atherosclerosis. Atherosclerosis is a cardiovascular remodeling as an adaptation to the shift of fat/water ratio in body.



The unicarbon compound transition system is tightly connected with the methyl group oxidation pathway ("detoxification pathway"). Glucose elevation leads to the synthesis of amyloid protein, changes of proton flows and neuromediator synthesis and accumulation in neurons.

CLINICAL INVESTIGATION OF NEURO-DEGENERATIVE DISEASES from WWMA AG



Full recovery Apparent improvement Treatment continue No effect



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DIABETES MELLITUS TYPE 2

FUNDAMENTAL and APLIED RESEARCH by WWMA AG

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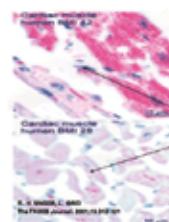
From 1985 to present the number of patients with diabetes increased by 6 times and amounted to 177 million. Major share of persons (80-95%) are patients with type 2 diabetes mellitus (DM). About 200 million people have impaired glucose tolerance (IGT), which is regarded as before stage type 2 diabetes. To the 2025 more than 300 million people are expected to be suffering DM.

Numbers of People with Diabetes 2000 - 2025



Particularly large increase in the incidence diabetes (an average of 170%) is expected in the population of Asia, Africa and Latin America. Prevalence of metabolic syndrome in the general population is ranging from 14 to 24% with IGT, while in DM - 80%.

Metabolic syndrome is an insulin resistance condition. Etiology is fat accumulation in extra-adipose tissues and abdominal visceral adipose tissue. Metabolic syndrome is a basis for hyperinsulinemia, diabetes mellitus type 2 and cardiovascular disease.



Oil red O (lipids) painting of heart muscle cells

Cardiac muscle autopsy material

obese 57 yrs old man (left)

(DM = 42)

lean 67 yrs old man (right)

(DM = 28)

Fat accumulation in muscle tissue is observed in both cases

Metabolic Syndrome



Type 2 Diabetes

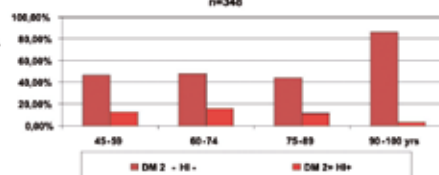
Coronary Heart Disease

Our lipid theory of aging suggests the same process of fatty acid accumulation in extra-adipose tissues as in metabolic syndrome and the same consequences expressed as age-associated diseases.

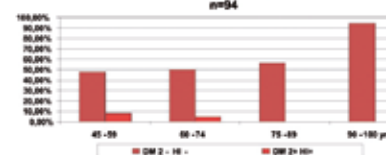
Diabetes mellitus type 2 is developing through the stages of insulin resistance, hyperinsulinemia, glucose tolerance impairment. We have investigated the prevalence of hyperinsulinemia combined with diabetes mellitus type 2 in men and women aged 45-100 yrs.

Women with DM 2+ HI+ have chance to live up to 100 yrs old while for men such diagnosis is a prognosis of deadline to 75 yrs old.

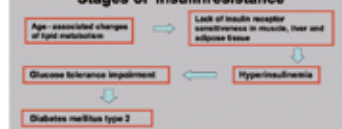
Prevalence of DM 2 + hyperinsulinemia in women aged 45-100 yrs n=348



Prevalence of DM 2 and hyperinsulinemia in men aged 45 - 100 yrs n=94



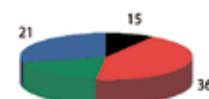
Stages of Insulin Resistance



Insulin resistance is a consequence of fat accumulation in extra-adipose tissues and a first step on the road to diabetes mellitus type 2.

The world first developed theory of systemic disorders and creation of the appropriate drugs and methods. Their application based on the convergence of methods of traditional Chinese medicine and on methods of nano-biopharmacology permits conduct prevention, treatment and rehabilitation of the most common diseases of mankind in the twenty-first century: pathologies of the cardiovascular system, diabetes and age associated diseases.

CLINICAL INVESTIGATION THE DIABETES TREATMENT by WWMA



Diabetes type 2
68 patients

Full recovery Apparent improvement Treatment continue No serious effects



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PROGRAMS for VIRAL HEPATITIS TREATMENT by WWMA AG

FUNDAMENTAL INVESTIGATION BY WWMA AG

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Aim of therapy programs The complete cure of patients treated for acute and chronic viral - inflammatory diseases - viral hepatitis including those accompanied by immunodeficiency.

General information about medicines used in therapy programs

Medicines XP/VH, GP/PQ, GP/BC are solutions of active natural components in isotonic aqueous solution of sodium chloride with additional composition of microelements. These solutions were produced according to a special technology and in high dilution.

Patients

- Patients with acute viral hepatitis, chronic viral hepatitis B, C, A, B+C were randomized for treatment either following a basic scheme or including application of a set of medicines in doses: 5 ml XP/VH/3 ml GP/BC/ 2 ml GP/PQ. These medicines were infused intramuscularly monthly according to certain schemes in the course of 24 months. The treatment was supplemented by detoxification as well as anti-inflammatory, hepatoprotective therapy. Number of patients: 88 (test group), 30 (control group) - men and women aged 18-73.

Therapy programs' efficiency

- Erythropoiesis normalization;
- Normalization of erythrocyte content; Reduction of manifestation up to complete elimination of chronic inflammation syndrome;
- Decrease of hepatocyte's cytolysis;
- Leveling of chronic fatigue and astheno-vegetative syndromes: increased tolerance of physical and psychoemotional loads, lower fatigability, optimism;
- Decreased intensity and manifestation of accompanying viral-inflammatory diseases up to complete normalization of health state;
- Stabilization and decreased manifestation of liver cirrhosis;
- Improved histological parameters of liver;
- Normalization of immune status indicators;
- Medicines proved to be highly efficient applied both in monotherapy (program) or in combination with traditional Chinese medicines;
- Program approach favours transition of infection from latent to active phase, thus medicines can be used for purposes of differential diagnostics of diseases following a short intensive course;
- Action mechanism of XP/VH medicine is realized on the level of perception, presentation and recognition of antigen, i.e. on the level of antigen-specific immune response;
- Program therapy efficiency is determined by the duration of viral-inflammatory process as well as by the initial state of immune system and depends on the initial pre-activation/pre-stimulation of organism and the prevailing cytokine profile which, in its turn, is determined by antigen's chemical properties, its dose, pathways of entry into organism and the main affecting biotope;
- XP/VH medicine has an expressed influence on regenerative processes of various types of cells including blood cells;
- GP/PQ and XP/VH medicines, being metabolites of biochemical processes in cells, optimize key regulatory processes on cell and organism levels.

CONCLUSION

Program therapy in the period 2001-2012 allowed fine-tuning of methods for treatment of: viral hepatitis B - efficient as to complete cure of 95 % of patients and viral hepatitis C - efficient as to complete cure of 75 % of patients.

At the moment, the number of patients completely cured has reached:

- acute forms of hepatitis A - 57 patients (treatment duration - up to 6 months);
- hepatitis B - more than 80 patients (treatment duration - up to 2 years);
- hepatitis C - 17 patients diagnosed with hepatitis up to 18 years ago, 16 patients diagnosed up to 25 years ago were cured completely; 4 patients diseased more than 30 years ago underwent rehabilitation and improved their health state significantly

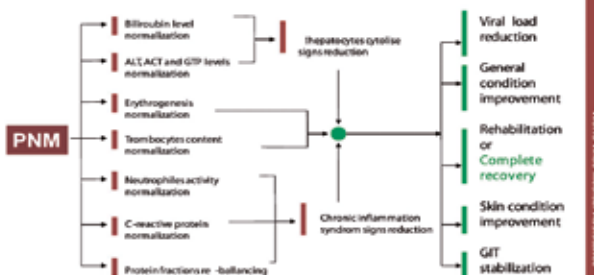
During the implementation of therapy programs clinical efficiency of certain therapy schemes in patients with viral hepatitis was assessed according to following data:

- Change in immune status;
- Mechanism of immunotropic activity;
- Medicines' curability and safety of application along with revelation and study of character, spectrum, frequency and intensity of possible side-effects and complications that become apparent in the course of certain modes of batching and therapy schemes.

During therapy implementation in 2001-2012 the following was established:

- No side-effects were found;
- Medicines used in the therapy programs do not interact negatively with other medicines;
- Medicines cause neither aggravation of previous chronic diseases nor intensification of manifestation of such diseases;
- Medicines have no toxic effects;
- No mutagenic, teratogenic, embryotoxic or cancerogenic effects were found;
- Medicines provoke no allergies and are well tolerated by patients.

PNM effect - hepatitis



CLINICAL CASES

Acute viral hepatitis A

Complete normalization of biochemical indicators, termination of hepatocyte cytolysis on either 2d or 3d week after the course.

Viral hepatitis B

In acute phase - faster normalisation of biochemical parameters indicating the level of hepatocyte cytolysis.

Normalisation of albumin and globulin ratio in blood.

Considerable decrease in viral replication up to negative PCR results.

Mix- viral hepatitis B and C

Termination or decrease in viral replication: hepatitis B after 4-7 months of therapy, hepatitis C - after 9-18 months.

Viral hepatitis C

Termination or decrease in replication of hepatitis C virus - from 6 to 15 months. Decreased clinical-biochemical activity of viral hepatitis C.



DNA methylation blocks the hepatitis viral genes expression. The blockade is forced by signals from the unicarbon transfer system that is connected with methyl group oxidation. Oxidation is stimulated by PNM injections.



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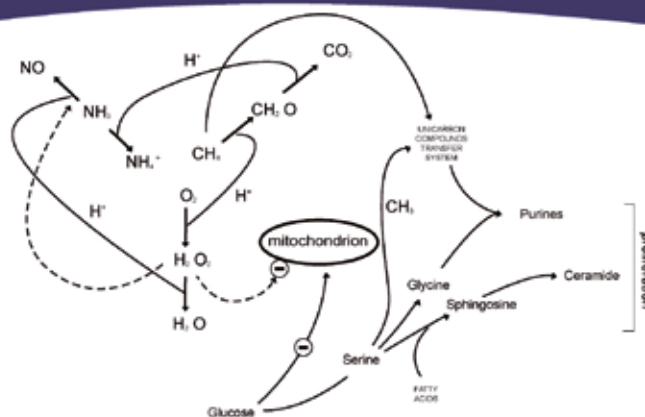


Integration of biomedicine and TCM in the treatment of cancer of different etiology.

Preparations and methods from WWMA AG, Switzerland

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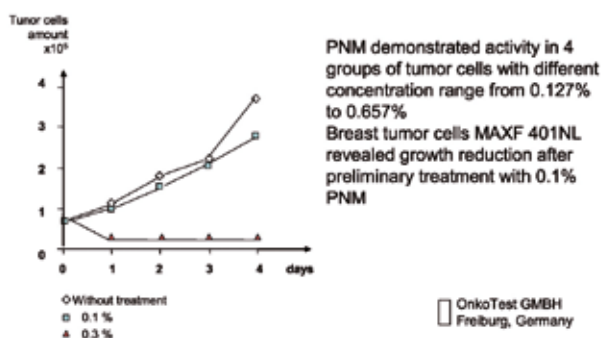
Our research is aimed at the development of nontoxic, multipurpose drug with antineoplastic effect on the majority of malignant neoplasms that will not have a destructive effect on normal cells and will take a possible viral disease origin into account.



Mechanism of PNM regulation of proliferation

Derived data about antineoplastic activity of the PNM drug and the results of researches in oxidation process in test animals indicate an active restrictive PNM action on neoplastic cells through changes in their metabolism.

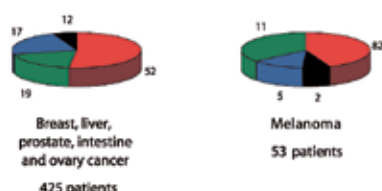
PNM effect - tumor cells



A maximum drug concentration ($3,6 \cdot 10^{-3}$) caused the death of the cells, concentration in the range from $3,6 \cdot 10^{-5}$ to $3,6 \cdot 10^{-8}$ caused growth inhibition of neoplastic cells.

Data of PNM clinical investigations – cancer diseases

In series of researches which have been carried out by the leading medical centers and clinic of Moscow and still proceed nowadays, it was shown, that in patients with breast, intestines, ovary tumors, skin melanoma, (in total 488 patients, age from 26 to 81 years):



Full recovery Apparent improvement Treatment continue lethal

The drug was tested in vitro in human cell cultures:

- rectum carcinoma (CXF 280, CXF 1103),
- stomach (GXF 1172),
- liver (LIXF 575),
- bladder (BXF 1218),
- lungs (LXFA 289 и LXFA 526),
- small cell carcinoma of lungs (LXFS 615, LXFS 650),
- large cell carcinoma of lungs (LXFL 1647 и LXFL 529),
- squamous cell epithelial carcinoma of head and neck (HNXF 536),
- adenocarcinoma of breast (MAXF 401, MAXF 1322, MAXF 1384),
- carcinoma and papillary serous adenocarcinoma of ovary (OVXF 550 и OVXF 899),
- adenocarcinoma of pancreas (PAXF 736),
- prostate (PRXF DU 145 и PRXF MRIH1579),
- melanoma (MEXF 1539, MEXF 514, MEXF 989),
- pleural mesothelioma (PXF 1118),
- clear-cell adenocarcinoma of kidney (RXF 631, RXF 944 LX и RXF423),
- malignant rhabdomyoma (SXF 627).



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FARMAFORESI ELETTRODICA®

Preparations and methods from WWMA AG, Switzerland

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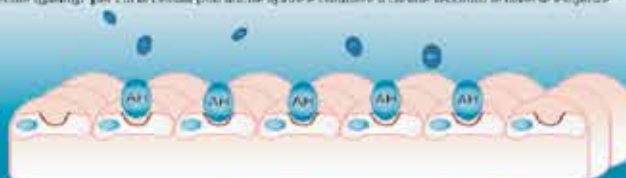
FARMAFORESI ELETTRODICA®

permette di raggiungere profondità inaccessibili ad altre tecniche senza invasione del circolo sistemico, mantenendo integra l'efficacia terapeutica. Per questo permette di trattare con successo le patologie più profonde e soprattutto quelle ritenute "impossibili" dalla scienza mondiale con i normali mezzi terapeutici. Solo grazie alle specifiche caratteristiche della FARMAFORESI ELETTRODICA® si ottengono contemporaneamente gli effetti elettromeccanico e biologico.

DIMENSIONE: 320x150x280 mm | **PESO:** 7 kg
VERSIONI PROPOSTE: Farmaforesi Trans Epidermal Barrier®
+ Aesthetic | Farmaforesi Trans Epidermal Barrier® Physio



Rappresentazione della dinamica molecolare del recettore H1 e del concetto di Occupazione Recettoriale (OR). Il dispositivo, tramite frequenze elettriche particolari, crea le condizioni per l'apertura e la chiusura delle molecole (gate) per cui la cellula può anche aprire e chiudere il canale secondo le diverse esigenze.



FARMAFORESI ELETTRODICA® è unica in caso di

- Osteopenia mascellare-mandibolare
- Fratture chiuse/ancora formazione calli ossei
- Capsulite adesiva
- Borsiti
- Fascite plantare
- Lesioni menisicali
- Strappi muscolari
- Artrosi d'anca
- M. di Gundeck
- Eritia discale cervicale
- Tunnel carpale - tarsale
- Dito a scatto (tenosinovite stenizzante)
- Lesione tibiale
- Tendiniti calcificanti
- Distorsioni 2°-3° grado
- Artrosi temporo-mandibolare
- Formazioni calcifiche - spine calcaree
- M. di Dupuytren
- M. di De Quervain
- Neuroma di Morton



CASE REPORT (Fondazione S. Raffaele Cittadella della Carità di Taranto)
Frattura aperta gamba sinistra o frattura caviglia destra. Paziente sottoposta ad intervento di osteosintesi con posizionamento di fissatore esterno. Rilevata per un quadro di pseudoartrosi con interruzione dei normali processi di guarigione della frattura, iniziava terapia transdermica a ripetizione con neridronato. Alle dimissioni, dopo 10 giorni di terapia la paziente era in grado di de-

TRATTAMENTI IN MEDICINA ESTETICA

Trattamenti decisamente importanti si ottengono soprattutto in medicina estetica con la rivoluzione fatta di fattori farmacologici attivati sempre "topically", superficiali e localizzati.



CAMPI DI APPLICAZIONE

Può permettere nuovi approcci e sviluppare l'azione terapeutica in medicina estetica e in terapia antipaga. Soprattutto in medicina rigenerativa, stimolando i fibroblasti mediante un apporto di sostanze essenziali, le cellule staminali mediante supplementazione, oppure con l'uso di coenzimi plasmatici e di fibroblasti, con l'effetto di ricostruire o riparare i tessuti lesi, ottenendo il ripristino delle funzioni anatomiche perdute. Trova applicazione in quasi tutte le parti del corpo. Trova applicazione in quasi tutte le parti del corpo. Trova applicazione in quasi tutte le forme patologiche non trattabili con altre metodiche.



Trova applicazione in caso di:

- invecchiamento viso-sottolineo del collo
- rughe, iperiperpigmentazioni cutanee, telangiectasie, acne, adiposità localizzata
- cellulite, smagliature, strie rosse, sanguisughe
- inestetismi degli occhi, grasso nei sopraccigli
- palpebre rilassate, zampe di gallina, accelerata caduta di capelli, insufficienza anodale
- ipertrofici, inestetismi post-intervento, cheloidi post-intervento, inestetismi



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